« Quantum Computations» Module



Laboratoire Junia Quantique

Introduction

I. Lefebvre

21 Janvier 2026

Présentations (in order of appearance)



- Gabriel CHÊNEVERT - Module coordinator

  Crypto , Quantum Computing



- Isabelle LEFEBVRE - In charge of

  Theoretical electronic structure : crystals, surfaces, nano …(CNRS)

- Jean-François ROBILLARD (Directeur Adjoint)

  Thermoelectricity (heat propagation in nanostructured materials)

- Samuel DELEPLANQUE

  Algorithmic & Optimization problems









WHY THIS MODULE?

Why should you learn this?

- An engineer at the beginning of the 21st century can't be unaware.
- Knowledge directly useful for a profession by combining the qualities of computer scientist, engineer and quantum physicist.
- A great need for engineers , informaticians knowing a bit of quantum physics
- Knowledge of the possibilities presented by quantum computers can be used very quickly, whatever the profession.

What you should understand

- A Quantum computer IS NOT a very big Classical computer
- Which kind of problem are addressed
- What kind of results can be expected
- What are the technical constraints due to quantum aspect
- Be able to follow current events and give an informed opinion (more than bit/qubit)
- Understand programs and realize some simple computations

Not too early

Access Online Quantum Computer

a real quantum device or simulator
accessed remotely through the internet via quantum cloud platforms.
These platforms often come with a user-friendly interface, development environments, and tools to run and visualize

Exemples

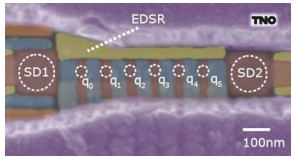


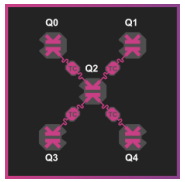
Free

Quantum Inspire - By QuTech

TU Delft TNO

Free





Amazon Braket

IONQ

rigetti

OQC

Pay-per-shot

D-Wave

2025 : Free → Pay

Overview

- Introduction
- Quantum mechanics (Lecture + Tutorials)
- Quantum gates (Lecture)
- Quantum algorithm (Lecture + Practical sessions on IBMQ)
  - Shor algorithm
  - Grover algorithm
- Physical qubits (Lecture at IEMN) + IEMN Lab tour



- Quantum optimization (Lecture + Practical sessions + Pasqal Invited Talk)





ZRR : Zone à Régime Restrictif

Interdiction de pénétrer sans autorisation

Tout contrevenant s'expose aux peines prévues par l'article 413-7 du Code Pénal

Visite le 11 février 2026

Demande d'accès IEMN



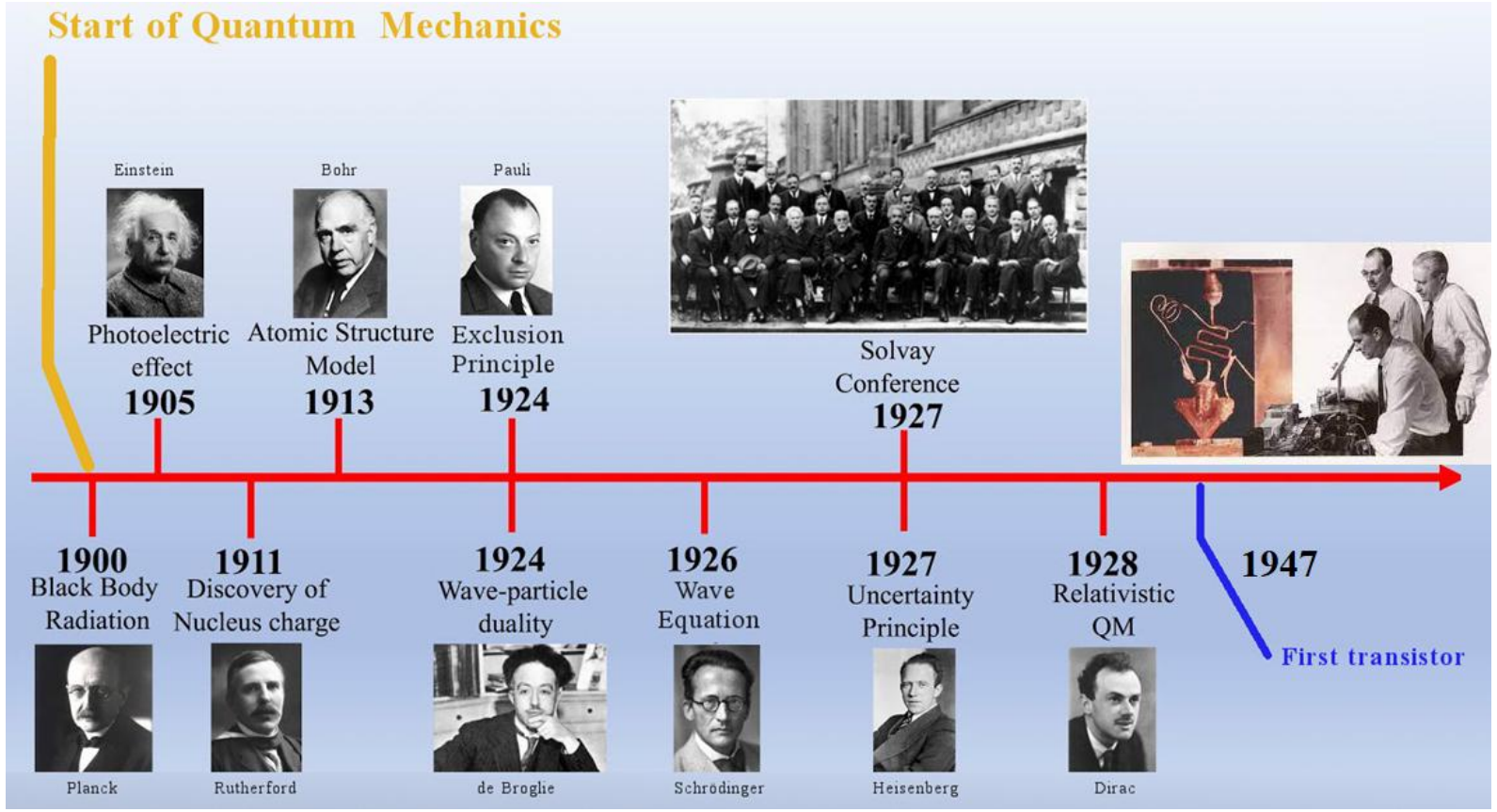
A quantum computer

Once upon a time…

A bit of history

Construction – First revolution

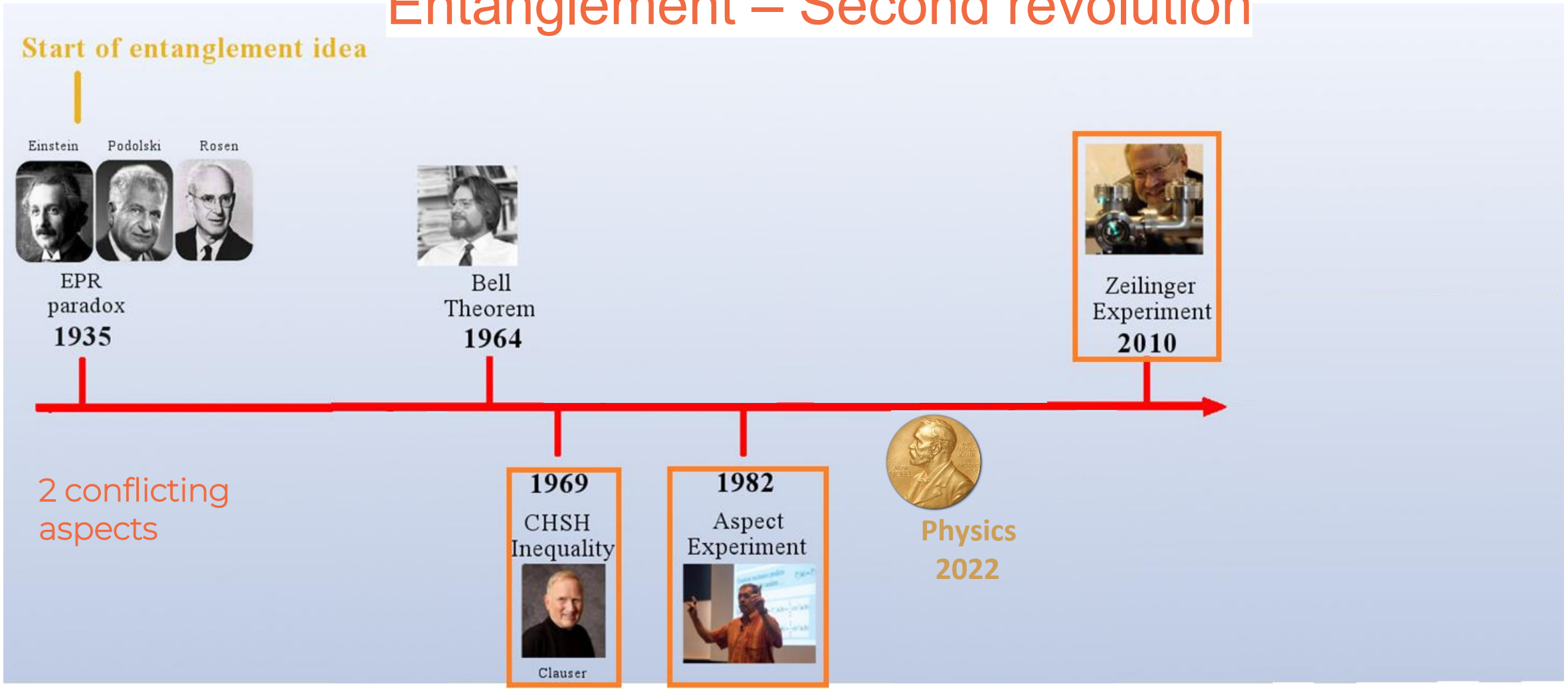
Discovery of concepts





Control of particle groups

Entanglement – Second revolution



Entanglement – Second revolution

Start of entanglement idea



EPR paradox **1935**



Bell Theorem **1964**



Zeilinger Experiment **2010**

1998 Chuang
First Q Comp. 2 qubits
molecules / NMR



2 conflicting aspects

1969 CHSH Inequality



Clauser

1982 Aspect Experiment

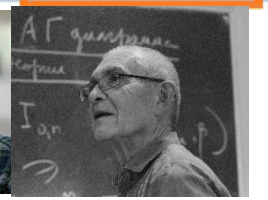




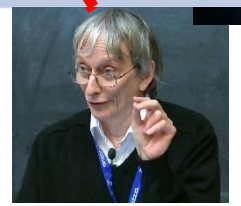
Physics 2022

1980 Yuri Manin - Paul Benioff
Quantum computer ideas/theory





1981 Feynmann à Endicott - Physics of Computation



1985 David Deutsch
Develops a Türing machine
Demonstrates universality

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1964



macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit
1984-85



Physics 2025



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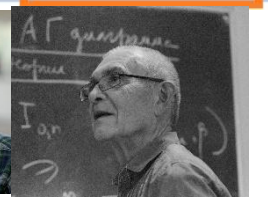




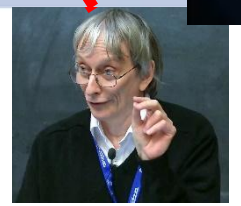
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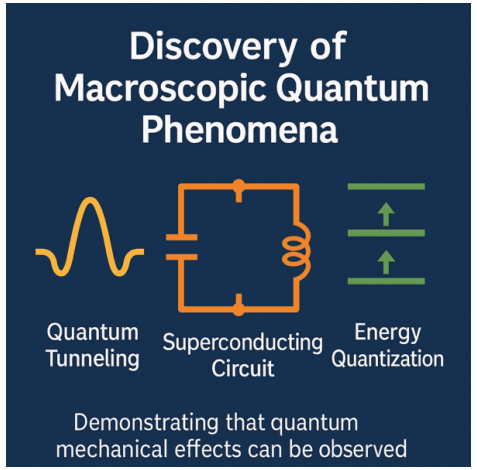




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See J-F. Robillard class

VOLUME 53, NUMBER 13 PHYSICAL REVIEW LETTERS 24 SEPTEMBER 1984

Resonant Activation from the Zero-Voltage State of a Current-Biased Josephson Junction

Michel H. Devoret,[a] John M. Martinis, Daniel Esteve,[a] and John Clarke
Department of Physics, University of California, Berkeley, California 94720, and Materials and Molecular Research Division, Lawrence Berkeley Laboratory, Berkeley, California 94720
(Received 26 July 1984)

Invitation to a seminar



Daniel Estève

Membre de l'Académie des Sciences et chercheur au Quantronics, Service de Physique de l'Etat Condensé (CEA Saclay).

The 2025 Nobel Prize in Physics: launching the area of superconducting quantum electrical circuits



Multiple-choice question

Jeudi 5 février 2026, à 10h00 Amphithéâtre de l'IEMN

Inscription : https://evento.renater.fr/survey/seminaire-daniel-esteve-doy51a5g
Visio : https://univ-lille-fr.zoom.us/j/95778371280?pwd=YnhwNMQWbJcJLHtcl9OKyZdWtJYPzn.1

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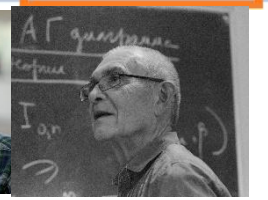




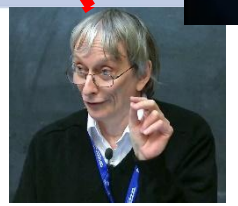
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2015–2018 | Early Commercialization

IBM launched cloud-based quantum access
D-Wave introduced quantum annealing for optimizion problems
Startups like Rigetti and IonQ entered the market

2019–2021 | Cloud & Ecosystem Expansion

Google achieved the quantum supremacy milestone
Amazon Braket enabled multi-vendor quantum access via the cloud
Microsoft focused on quantum software, tooling, and developer ecosystems

2022–2024 | Hardware Scaling & Enterprise Pilots

IBM crossed 1,000+ qubit systems
IonQ and Quantinuum advanced trapped-ion architectures
Priority: Error mitigation and real-world use cases.

2025–2026 | Utility-Driven Quantum Era

Quantum computing targets measurable advantage in

- Drug discovery & molecular simulation
- Financial risk modeling & portfolio optimition
- Logistics, energy systems & climate modeling
- Cryptography & cybersecurity

2026–2030 | Future Outlook

- Quantum computing shifts from experimentation to measurable advantage
- Cloud-first access accelerates enterprise adoption
- AI + Quantum convergence unlocks complex optimization
- Talent, algorithms, and error correction become key differentiators.

Leading Quantum Computing Players

IBM Quantum – Supercoinducting qubits & enterprise quantum cloud

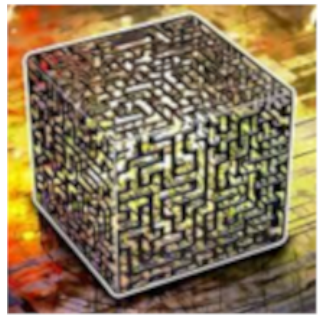
DATA INTELLIGENCE
January 2026

Computer types

Quantum digital ⟵ **Quantum analogic** **Classical**

Gate based **Annealing** **Simulator** **Emulator**

Universal **FTQC (Fault-Tolerant Quantum Computer)** **NISQ (Noisy Intermediate Scale Quantum)**



PsiQuantum

IBM

rigetti

IQM

QUANTINUUM

XANADU

Microsoft

IONQ

OQC

QUANDELA

ALICE & BOB

D-Wave The Quantum Computing Company

Pasqal

IQuEra

EVIDEN

IBM

FUJITSU

Google

Microsoft

aws

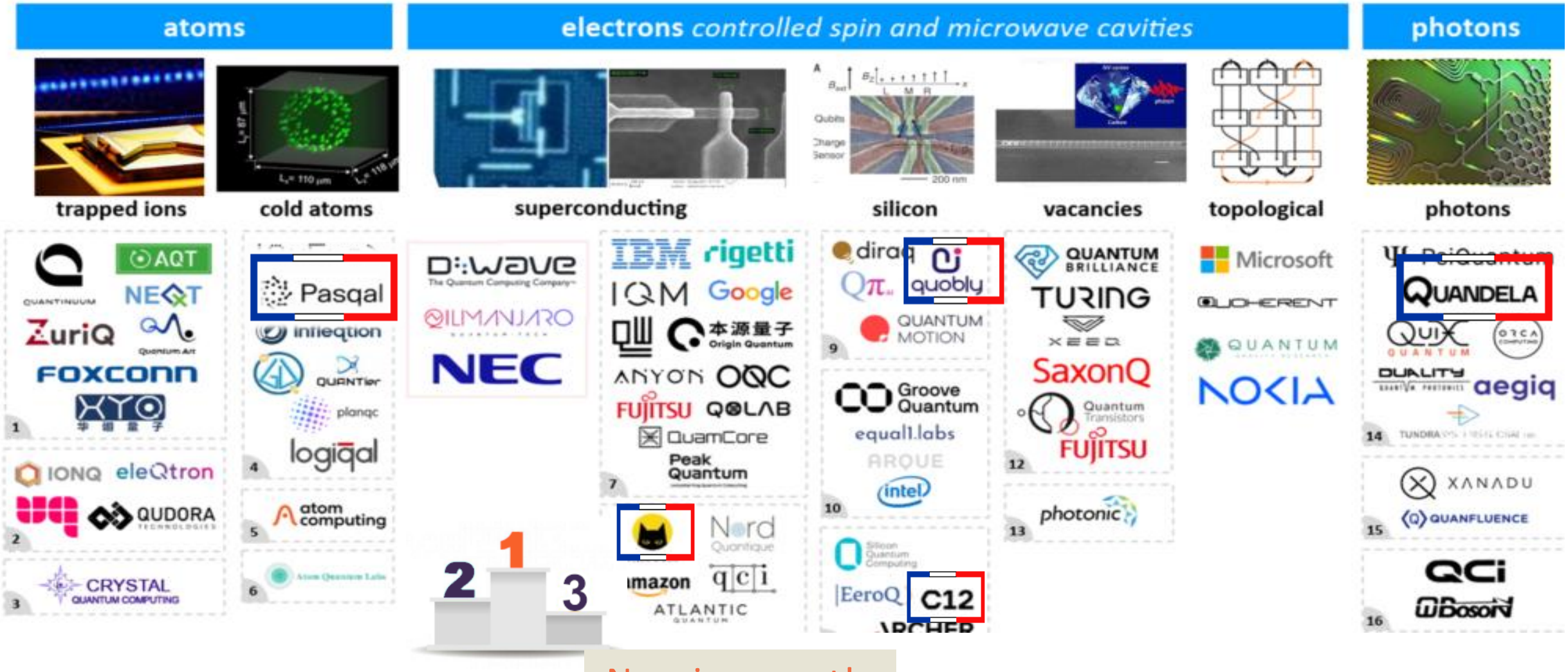
Pasqal

C12

QUANDELA

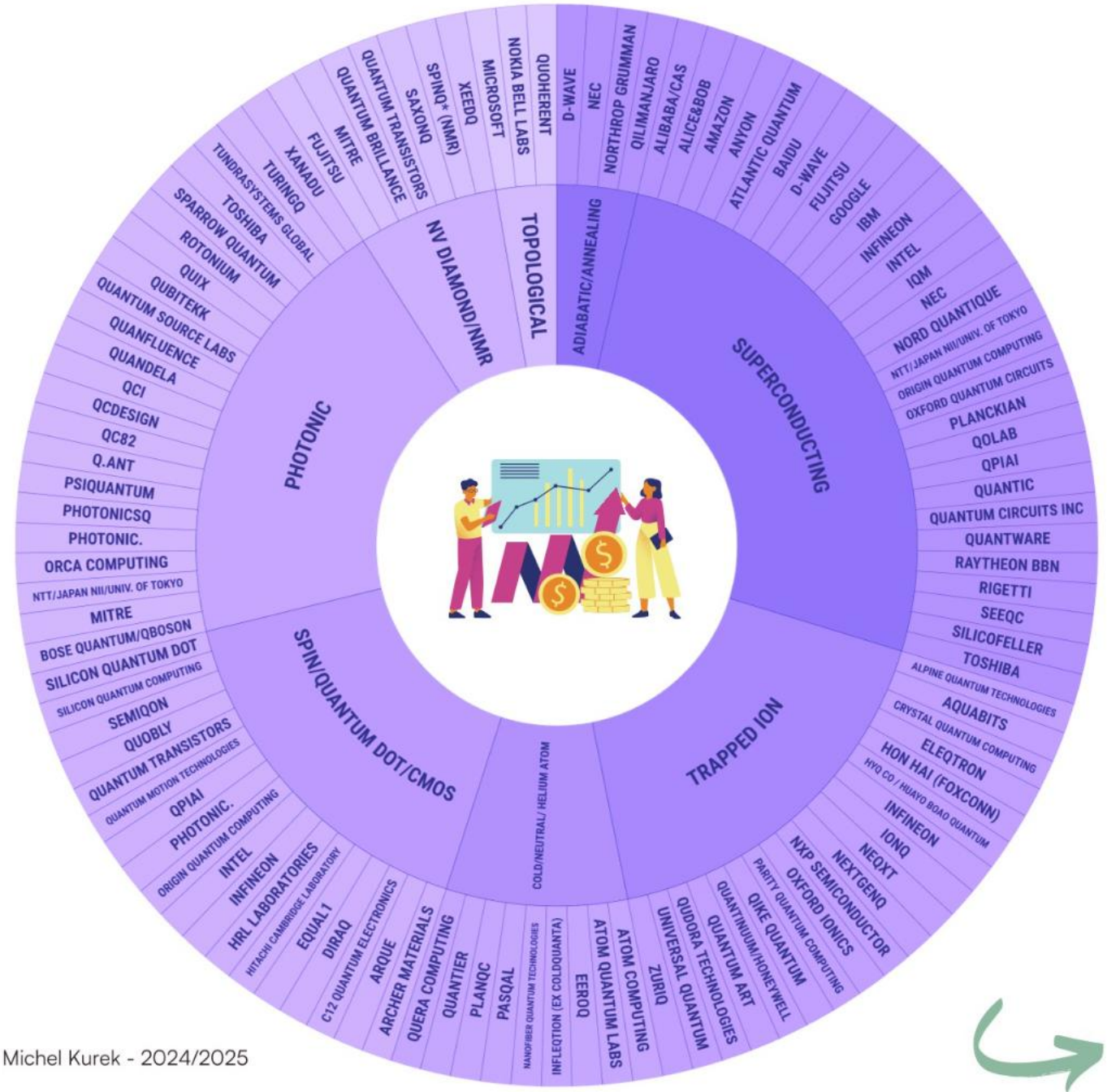
ALICE & BOB

Various technologies



No winner yet!

Inspiré de « Understanding quantum technologies 2024 » - O. Ezratty


QBIT MODALITIES - CORPORATES / STARTUPS /PRIVATE RESEARCH
TOPOLOGICAL
NV DIAMOND/NMR
PHOTONIC
SPIN/QUANTUM DOT/CMOS
COLD/NEUTRAL/ HELIUM ATOM
TRAPPED ION
SUPERCONDUCTING
ADIABATIC/ANNEALING
QUOHERENT
NOKIA BELL LABS
MICROSOFT
XEEDQ
SPINQ* (NMR)
SAXONQ
QUANTUM TRANSISTORS
QUANTUM BRILLANCE
MITRE
FUJITSU
XANADU
TURINGQ
TUNDRASYSTEMS GLOBAL
TOSHIBA
SPARROW QUANTUM
ROTONIUM
QUIX
QUBITEKK
QUANTUM SOURCE LABS
QUANFLUENCE
QUANDELA
QCI
QCDESIGN
QC82
Q.ANT
PSIQUANTUM
PHOTONICSQ
PHOTONIC.
ORCA COMPUTING
NTT/JAPAN NII/UNIV. OF TOKYO
MITRE
BOSE QUANTUM/QBOSON
SILICON QUANTUM DOT
SILICON QUANTUM COMPUTING
SEMIQON
QUOBLY
QUANTUM TRANSISTORS
QUANTUM MOTION TECHNOLOGIES
QPIAI
PHOTONIC.
ORIGIN QUANTUM COMPUTING
INTEL
INFINEON
HRL LABORATORIES
HITACHI CAMBRIDGE LABORATORY
EQUAL1
DIRAQ
C12 QUANTUM ELECTRONICS
ARQUE
ARCHER MATERIALS
QUERA COMPUTING
QUANTIER
PLANQC
PASQAL
NANOFIBER QUANTUM TECHNOLOGIES
INFLEQTION (EX COLDQUANTA)
EEROQ
ATOM QUANTUM LABS
ATOM COMPUTING
ZURIQ
UNIVERSAL QUANTUM
QUDORA TECHNOLOGIES
QUANTUM ART
QUANTINUUM/HONEYWELL
QIKE QUANTUM
PARITY QUANTUM COMPUTING
OXFORD IONICS
NXP SEMICONDUCTOR
NEXTGENQ
NEQXT
IONQ
INFINEON
HYQ CO / HUAYO BOAO QUANTUM
HON HAI (FOXCONN)
ELEQTRON
CRYSTAL QUANTUM COMPUTING
AQUABITS
ALPINE QUANTUM TECHNOLOGIES
TOSHIBA
SILICOFELLER
SEEQC
RIGETTI
RAYTHEON BBN
QUANTWARE
QUANTUM CIRCUITS INC
QUANTIC
QPIAI
QOLAB
PLANCKIAN
OXFORD QUANTUM CIRCUITS
ORIGIN QUANTUM COMPUTING
NTT/JAPAN NII/UNIV. OF TOKYO
NORD QUANTIQUE
NEC
IQM
INTEL
INFINEON
IBM
GOOGLE
FUJITSU
D-WAVE
BAIDU
ATLANTIC QUANTUM
ANYON
AMAZON
ALICE&BOB
ALIBABA/CAS
QILIMANJARO
NORTHROP GRUMMAN
NEC
D-WAVE

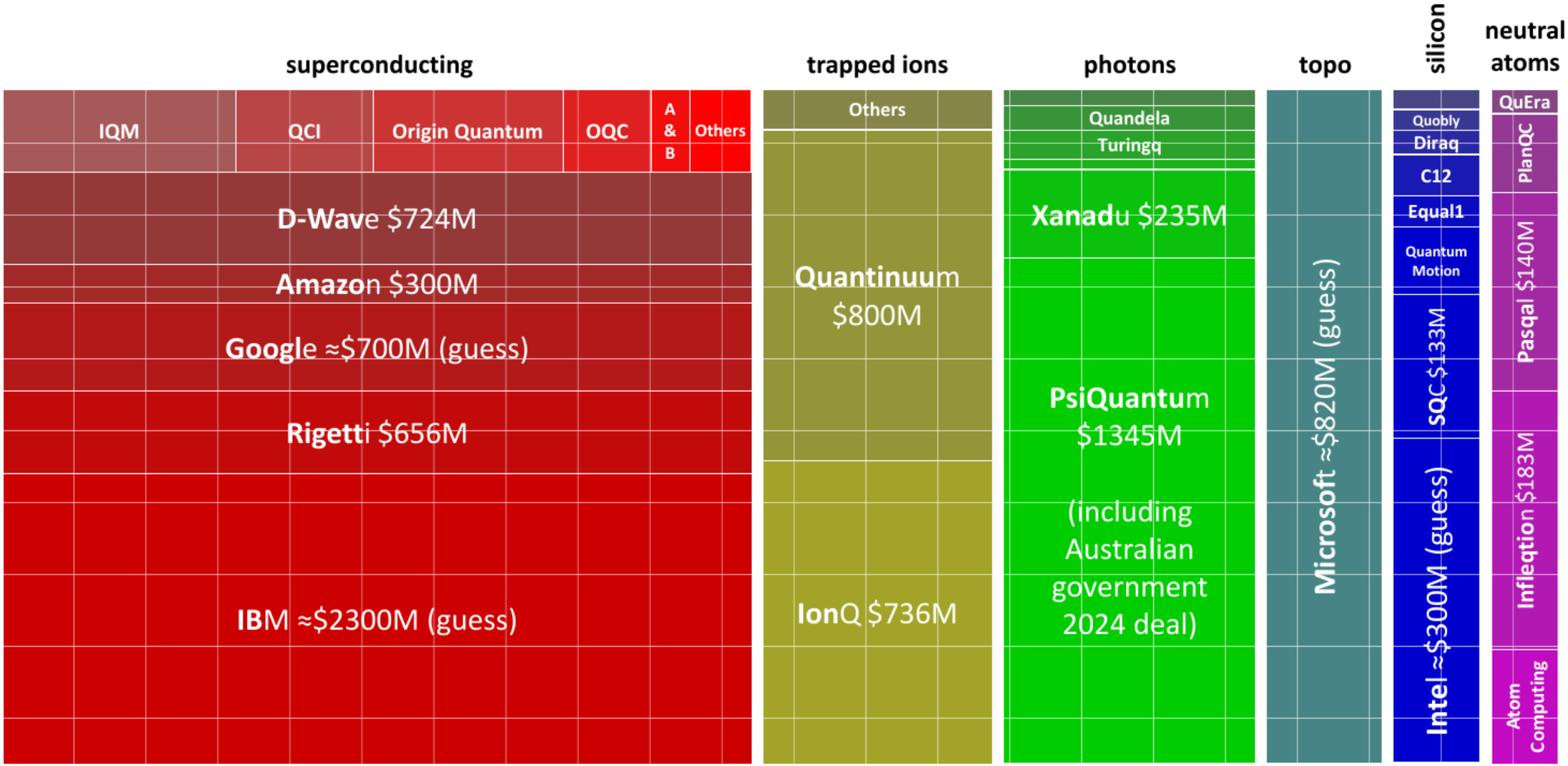
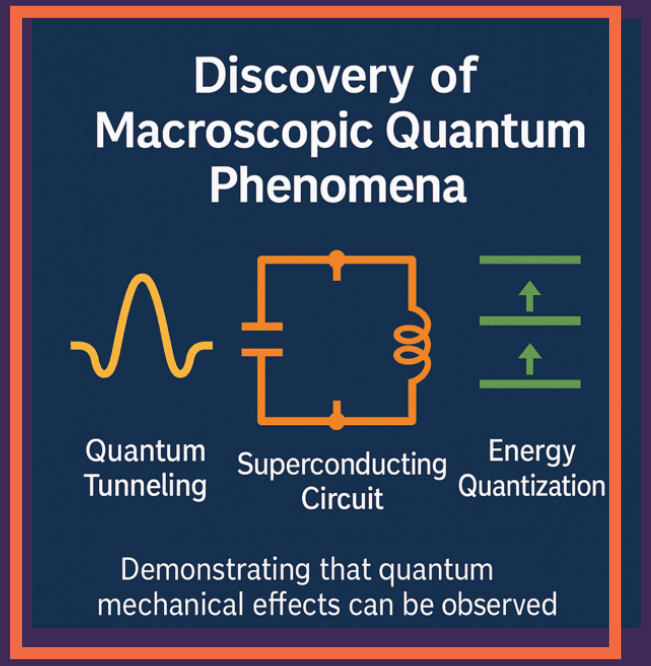
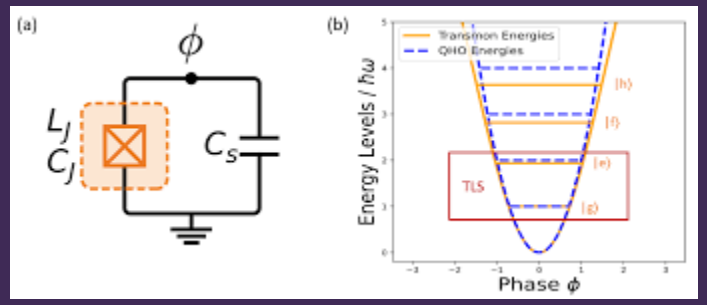


Figure 287: fuzzy logic assessment of industry investments per qubit type mixing capital investment for startups and internal investments for legacy companies is "life to date" (LTD). (cc) Olivier Ezratty, August 2024.



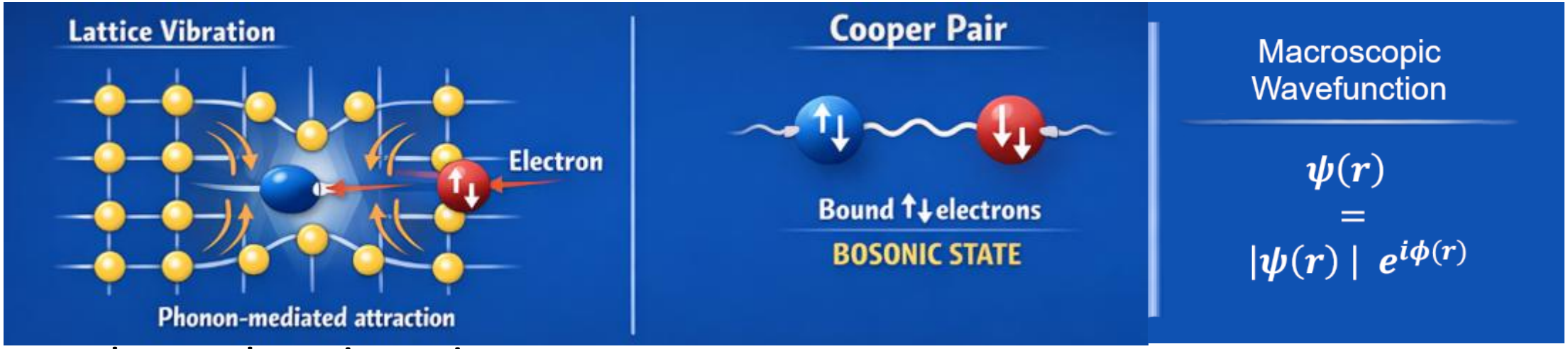


Back to the seminar: some words…



Cooper pairs / Superconductivity
Phase
Josephson junction
Quantum nonlinearity
Superconducting qubits

COOPER PAIR / BOSE EINSTEIN CONDENSATE / PHASE



electron–phonon interactions.

Two electrons form a **bound state**

$|\psi(r)|$: related to the density of Cooper pairs

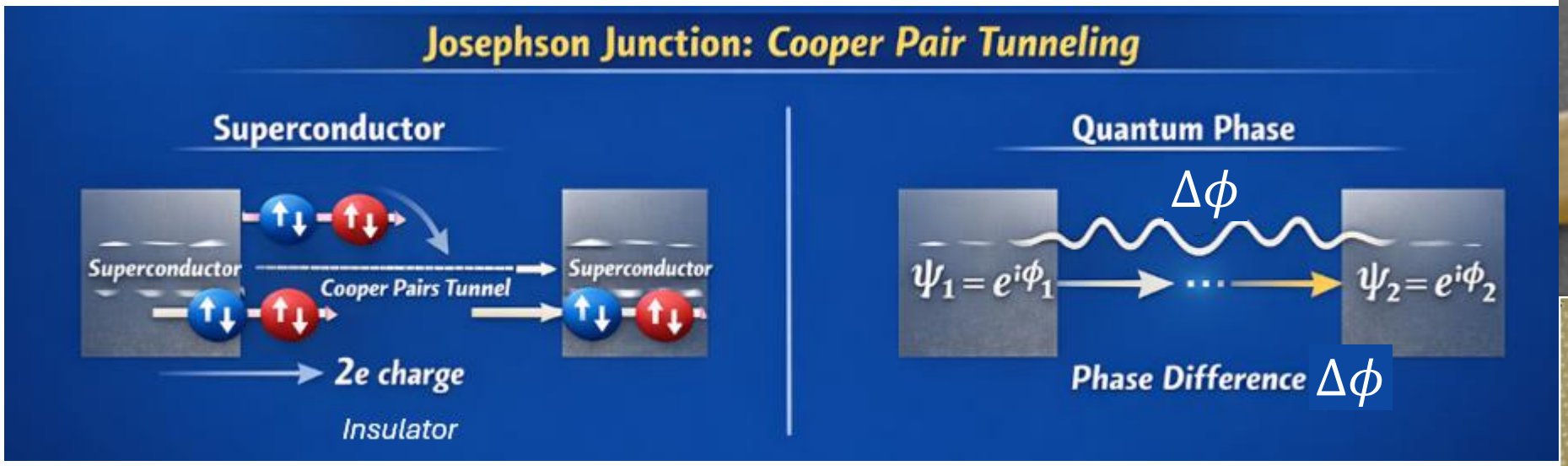
ϕ the phase of the Cooper pair wavefunction

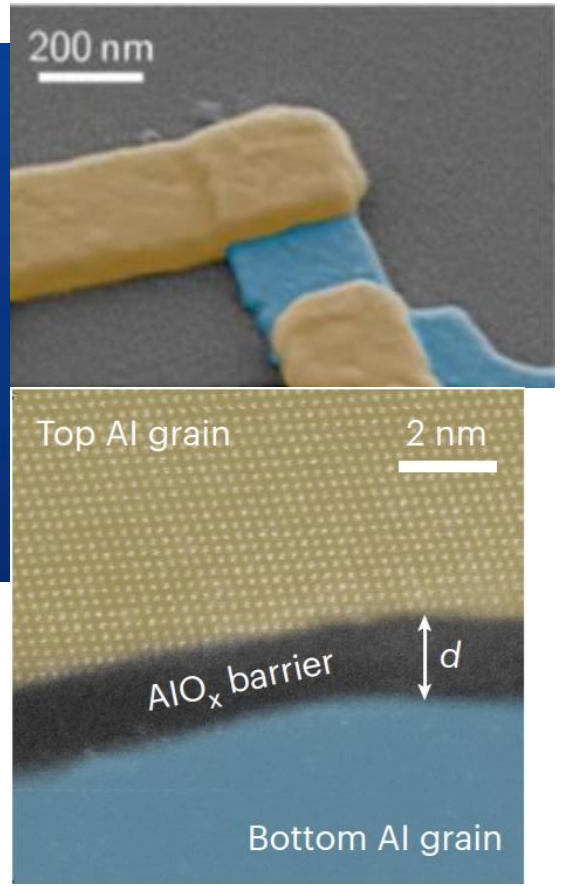
Cooled to **extremely low temperatures,**
all cooper pairs condense into a single macroscopique quantum state
described by a **single macroscopic wavefunction**:

ϕ is the key concept

JOSEPHSON JUNCTION

Control and exploit macroscopic quantum effects



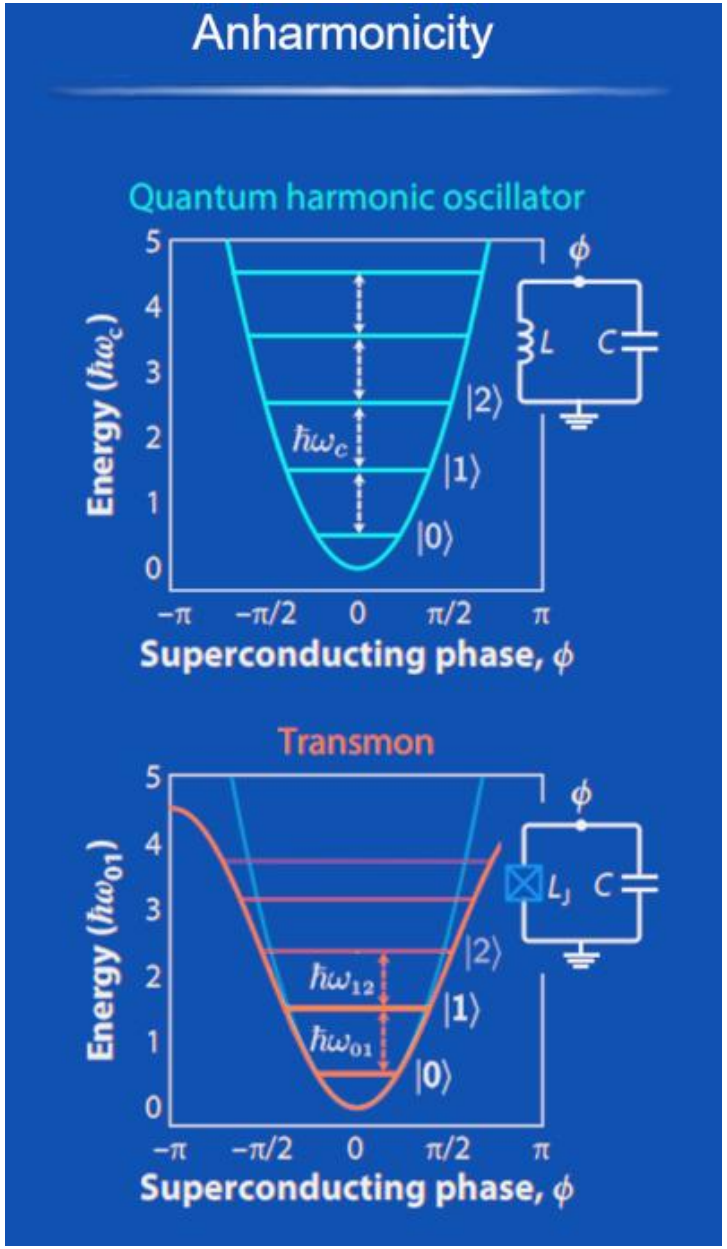


Josephson relation : $I = I_c \sin(\Delta\phi)$

Phase is a directly **measurable physical quantity**

QUANTUM NONLINEARITY

essential for creating and manipulating qubits.



Kjaergaard et al. Annu. Rev. Condens. Matter Phys. 2020. 11:369–95

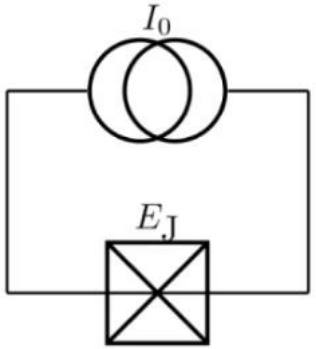
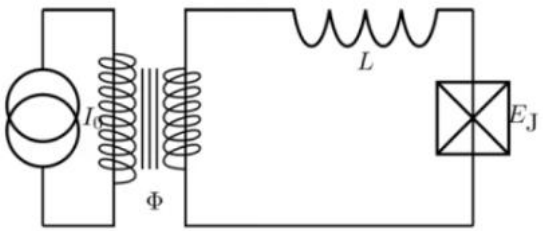
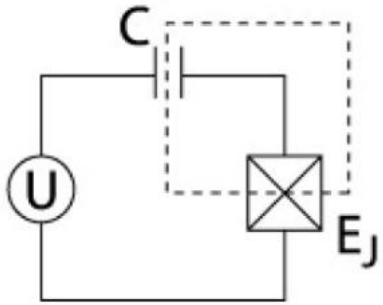
$$I = I_c \sin(\Delta\phi)$$

Potential energy

$$U(\phi) = -E_J \cos(\Delta\phi)$$

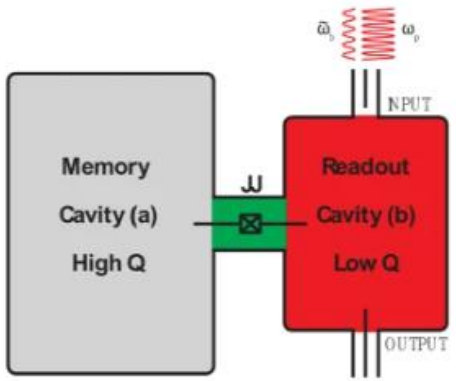
$$E_J = {}^{\hbar I_c}/_{2e}$$

QUBITS / JOSEHSON

	phase qubit	flux qubit	charge qubit - transmon
			
	Josephson junctions handle the qubit degree of liberty		
|0⟩ and |1⟩ qubits	two energy levels in a potential well	two superconducting current directions	two levels of charge of Cooper pairs
quantum gates	micro-waves	magnetic field	micro-waves
commercial vendors	**abandonned**	D-Wave The Quantum Computing Company, rigetti, bleximo, Alibaba, ATLANTIC QUANTUM, KILIMANJARO QUANTUM-TECH	IBM, Google, IQM, OQC

QUBITS / JOSEHSON

cat-qubits



Josephson junctions prepare, couple and correct the cat-qubits

|0⟩ and |1⟩ qubits	pairs of entangled microwave photons in a cavity
quantum gates	micro-waves



ALICE & BOB

qci amazon

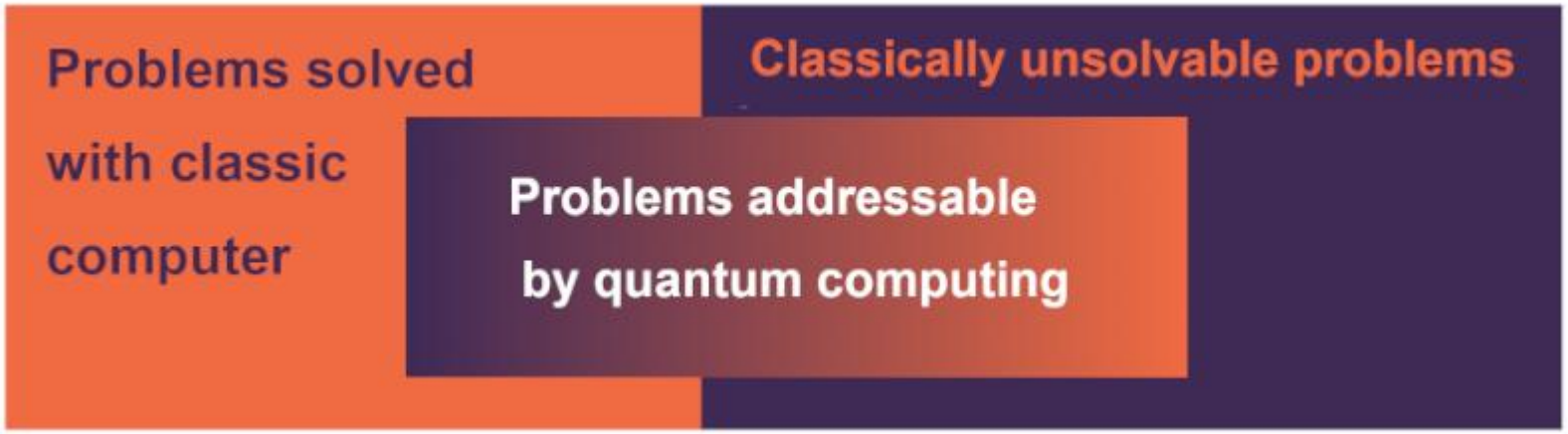
See you
on Thursday, February 5, 2026,
at 10:00 a.m. for the seminar.

Let's go back to the point



A quantum computer

A SUPER classical computer?



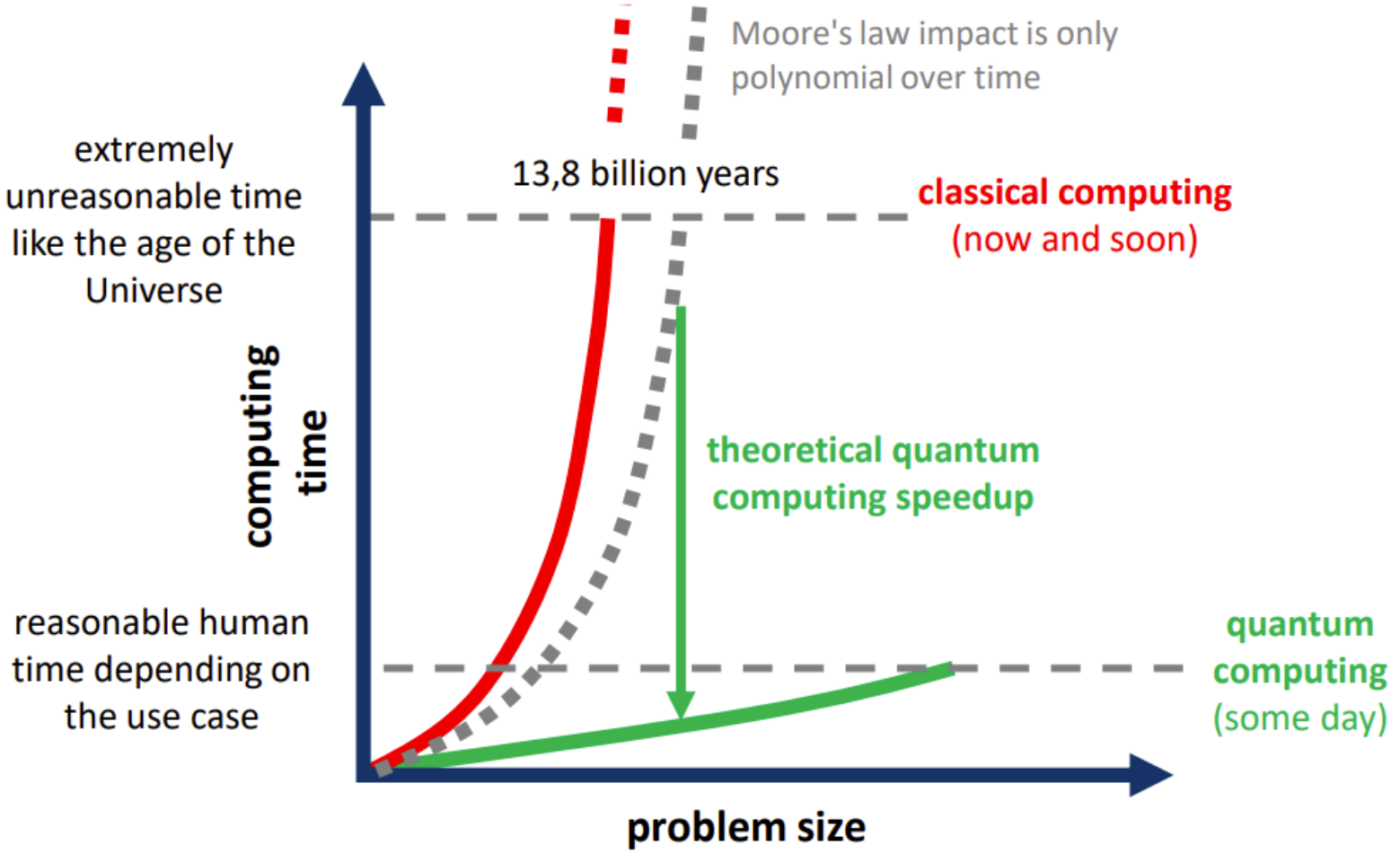
√ Faster √ Greater computing power

Δ For certain problems and certain computations

Thinking with quantum concepts
New paradigm



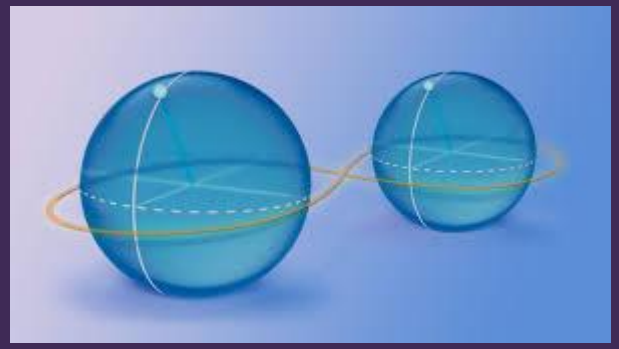
Cf Thomas KHUN



solving intractable / exponential problems in reasonable time

(cc) Olivier Ezratty, 2022

« Qubit » survey



Information

Qubit = quantum system
with a property that can take two values: « 0 » et « 1 »

Classical Bit



Quantum Qubit



Superposition



Mesure





Result



Probabilistic
Intrinsic randomness

with a **x%** chance

with a **(1-x)%** chance

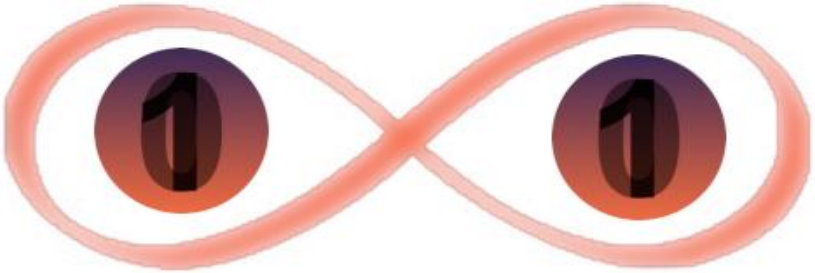
All possibilities



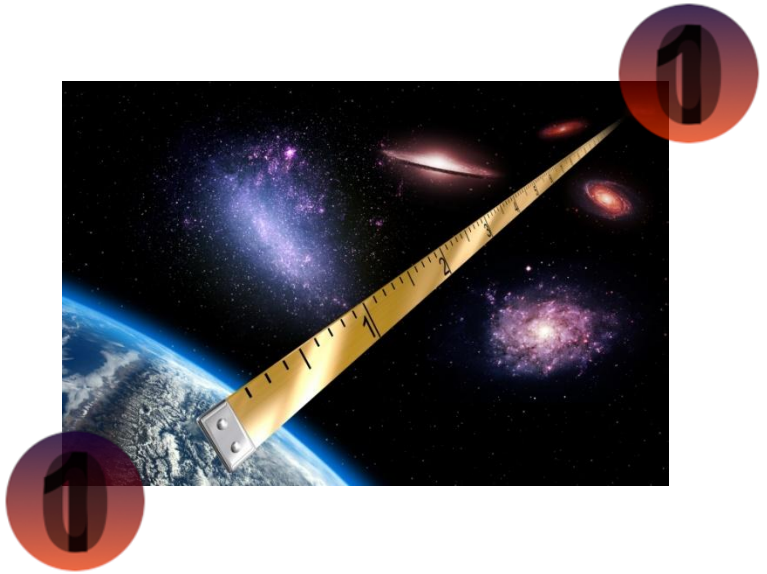
Entanglement

System of 2 ENTANGLED qubits



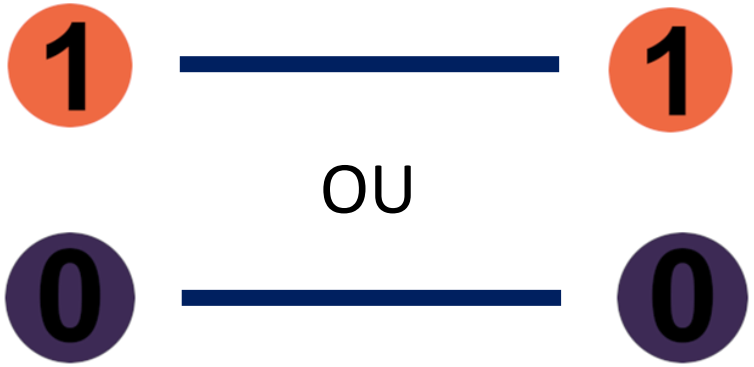






INDIVISIBLE WHOLE

If one of the two is measured, then the other **instantly** takes on the correlated value.

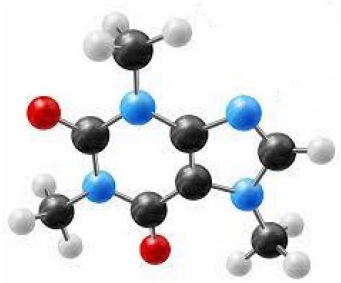


Computing power

Classical register		Quantum register	
Bits	Represent	Qubits	Represent
2	00	2	00
	or 01		AND 01
	or 10		AND 10
	or 11		AND 11
	4 possible sates		4 possible states
	one at a time		**at the same time**

Qubits	Classical Equivalent
2	4
3	8
4	16
5	32

Computing power



Caféine 24 atoms

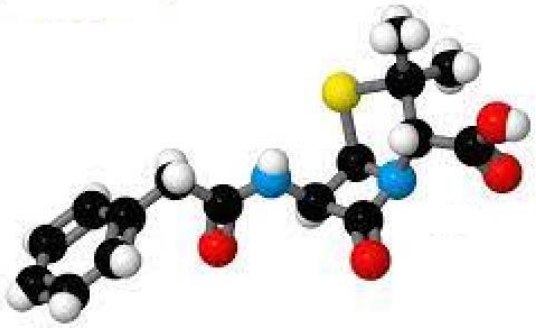


2^{160} Classical bits

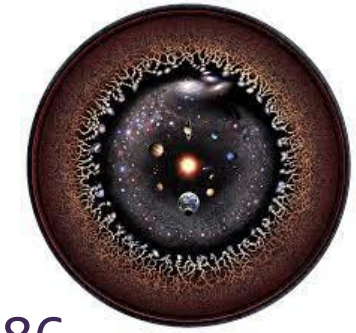
10% of the atoms on Earth



160 ideal qubits



Pénicilline 41 atoms



2^{286} Classical bits

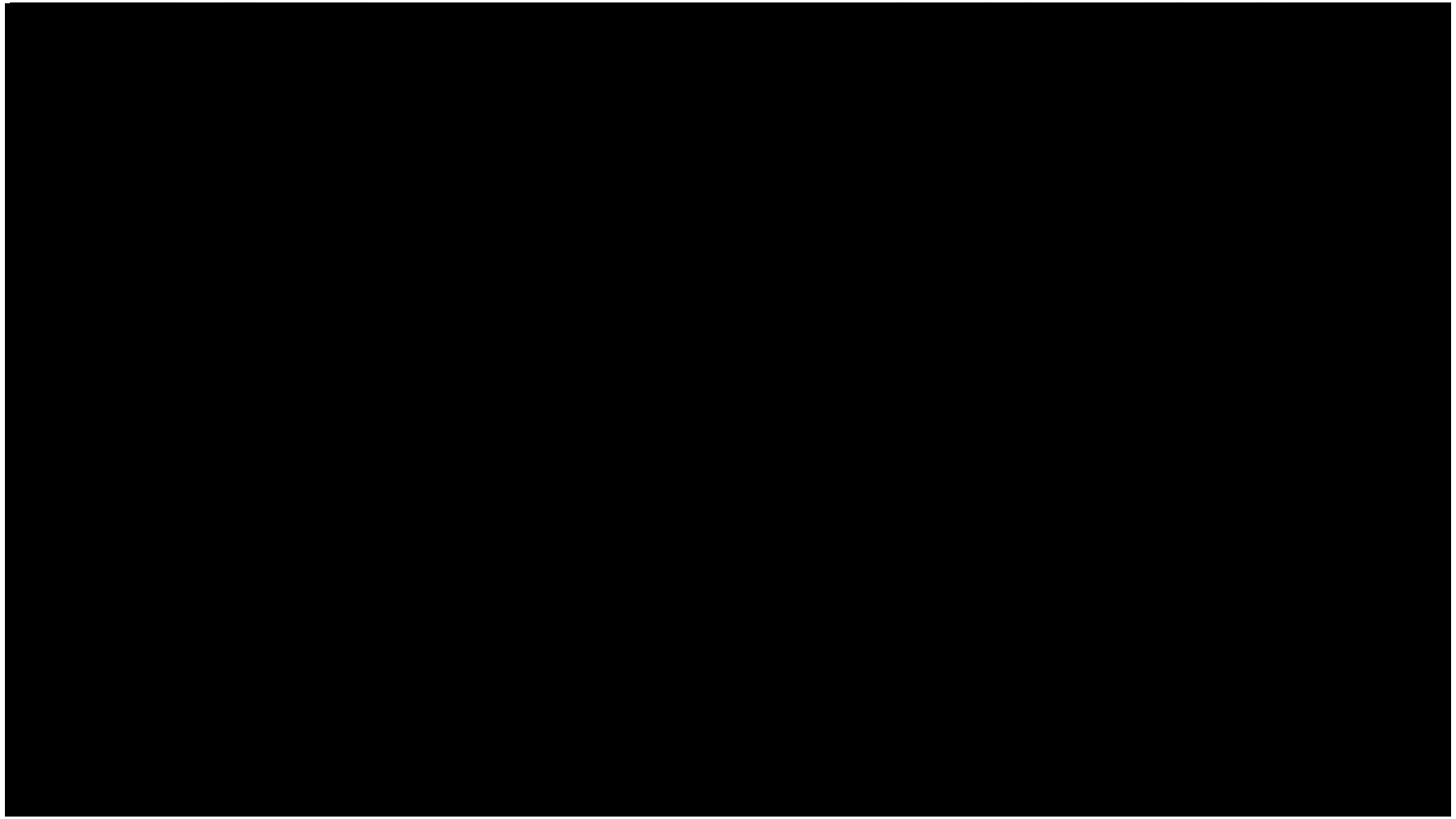
> of the observable universe



286 ideal qubits

Speed

Searching for the exit of a maze using a ball



Superposition de tous les chemins



Di Vincenzo criteria

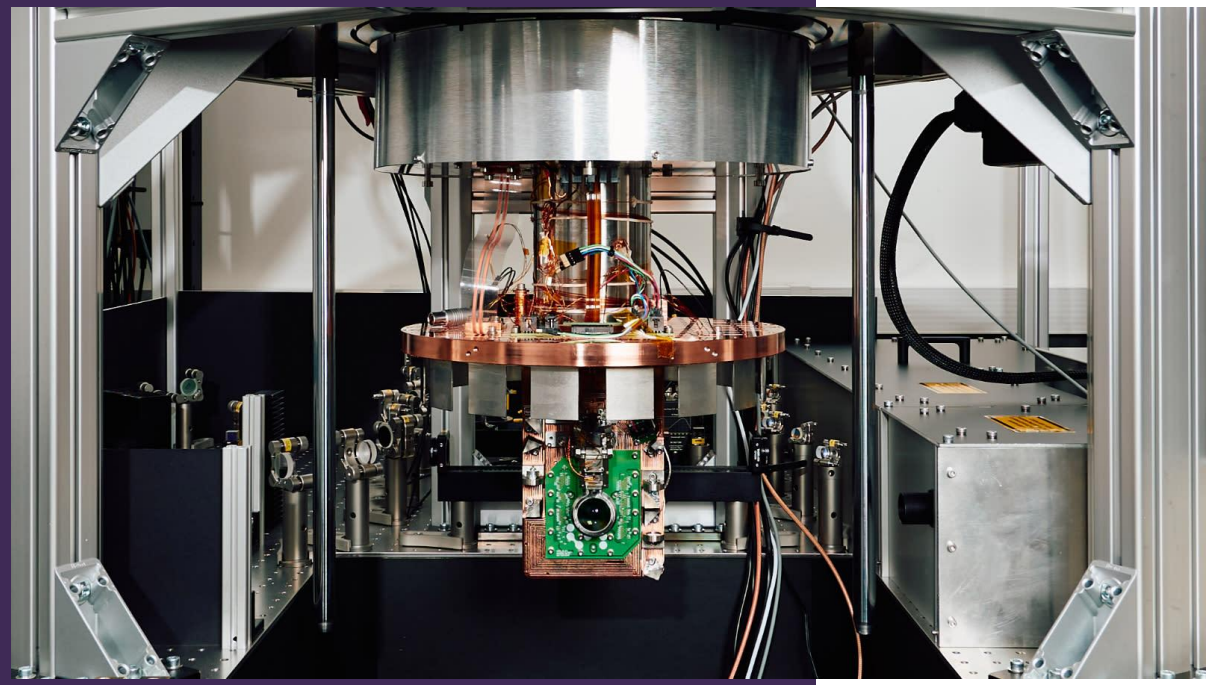
director of the Institute of Theoretical Nanoelectronics at the Peter Grünberg Institute in Jülich and Professor at the Institute for Quantum Information at RWTH Aachen University

In 1996, during his research at IBM

REQUIREMENTS FOR THE PHYSICAL IMPLEMENTATION OF QUANTUM COMPUTATION	
D1: Scalable qubits	Scalable physical system of well-defined, characterized qubits
D2: Initialization	Prepare a simple, fiducial input state
D3: Measurement	Measure the qubit state
D4: Universal gate set	Perform a universal set of gate operations with high fidelity
D5: Coherence	Robustly represent quantum information (long coherence times)
REQUIREMENTS FOR ROUTING QUANTUM INFORMATION	
D6: Interconversion	Ability to interconvert stationary and flying qubits
D7: Communication	Ability to transmit flying qubits faithfully between two locations

DiVincenzo, David (16 December 1996). "TOPICS IN QUANTUM COMPUTERS". Mesoscopic Electron Transport. arXiv:cond-mat/9612126

Machine reality



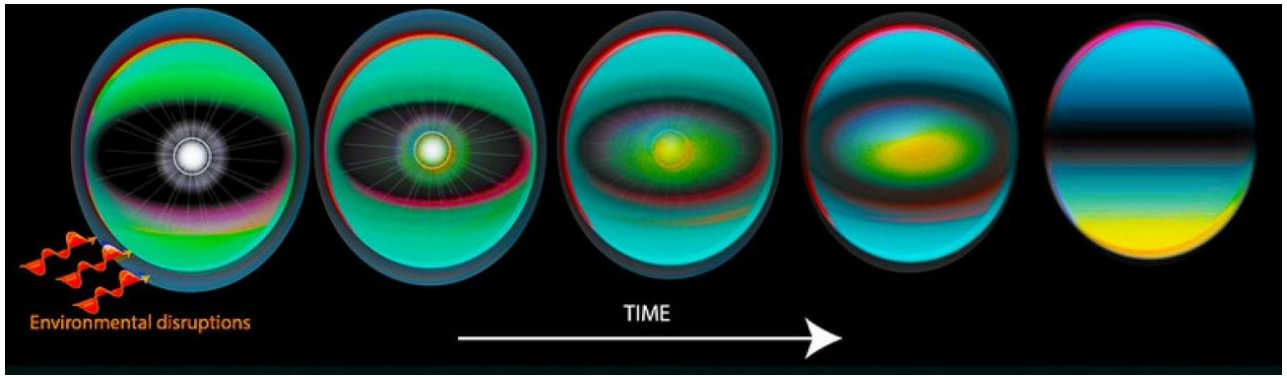
Oxford Ionics.

Decoherence

heat,
électromagnétic radiations,
croos-talk
and so on…

Extremely sensitive to small disturbances

Progressive **degradation** of a quantum system
under the influence of interactions with its environment



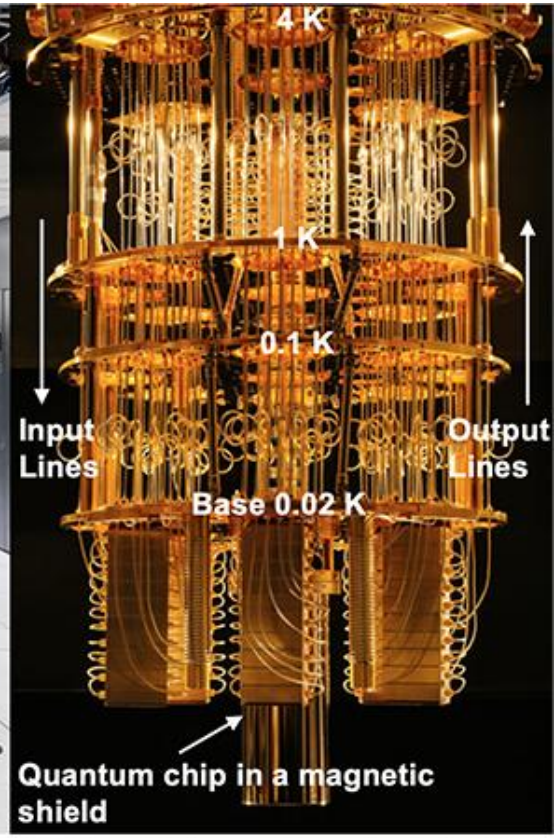
problem of scaling up

Need to manage **“errors”**

Isolate from disturbances



Dilution fridge setup: outside view



Dilution fridge setup: inside view

0,02K = -273,13°C (T cosmic radiation = -272°C)

Comparing quantum error suppression, error mitigation and error correction

Quantum error suppression	Quantum error mitigation	Quantum error correction
Attempts to prevent errors before they happen.	Attempts to reduce the effect of errors after they happen.	Attempts to detect and fix errors as they occur.
Quantum hardware is designed to be more resistant to noise.	Runs quantum circuits multiple times to estimate the error-free outcome.	Spreads a qubit's value across multiple physical qubits for redundancy.
Focuses on hardware-level noise reduction.	Focuses on post-processing.	Focuses on real-time error detection and correction.
Is the most basic level of handling errors.	Useful for today's noisy quantum devices.	More complex but necessary for fault-tolerant quantum computing.

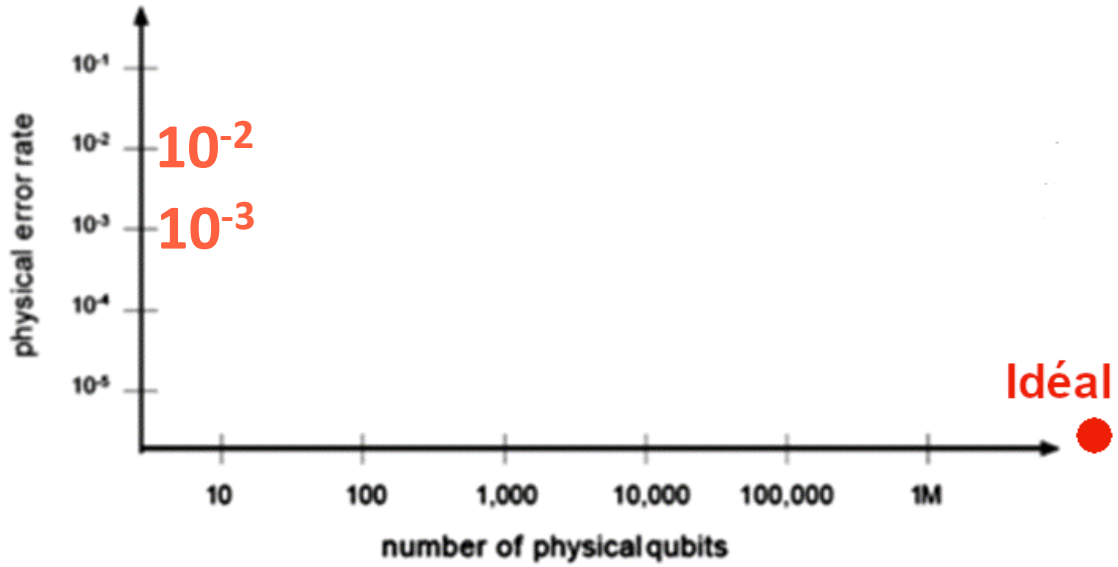
Cryogénie / isolement



1000 qubits physiques / 1 logique

- Number of qubits

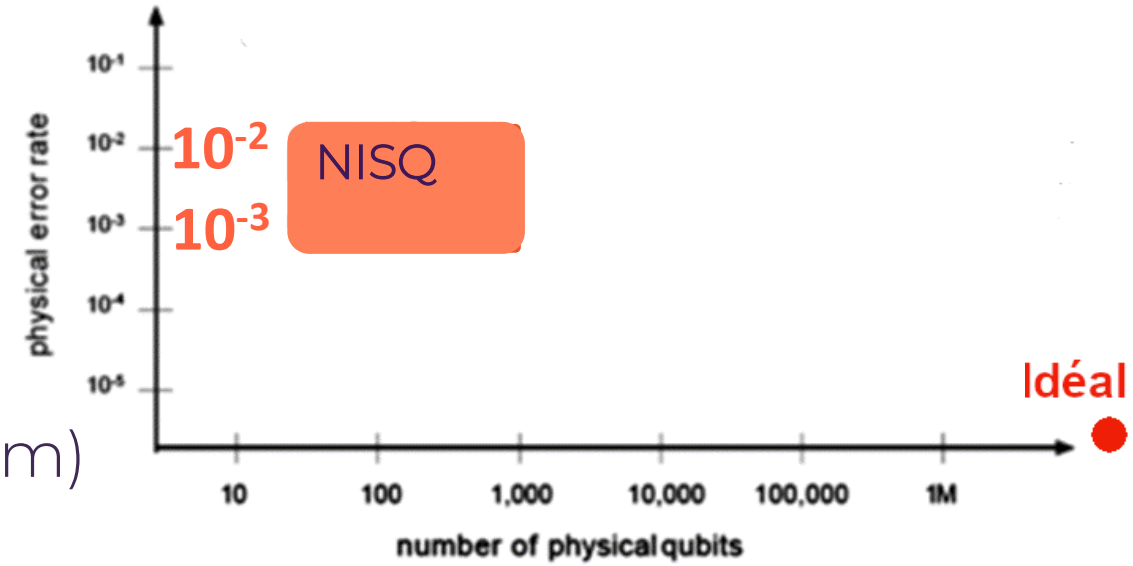
Universal quantum computer does not exist yet, but



- Number of qubits

Improving quality and
"dealing with" defects

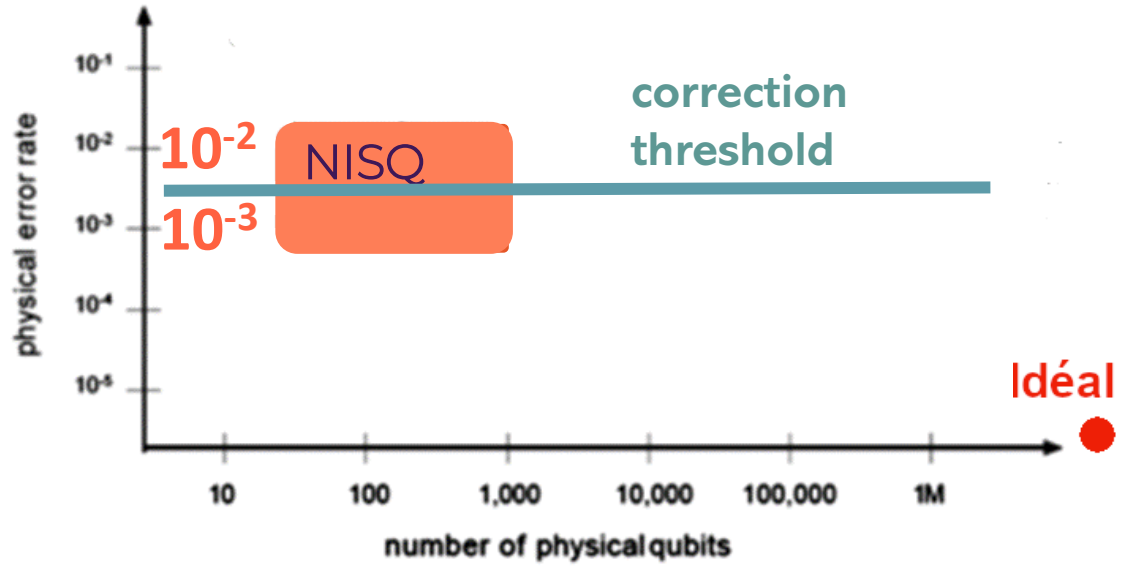
NISQ
(Nearly Intermediate Scale Quantum)



large enough to perform non-trivial quantum computations
errors limit the complexity and accuracy of the computations

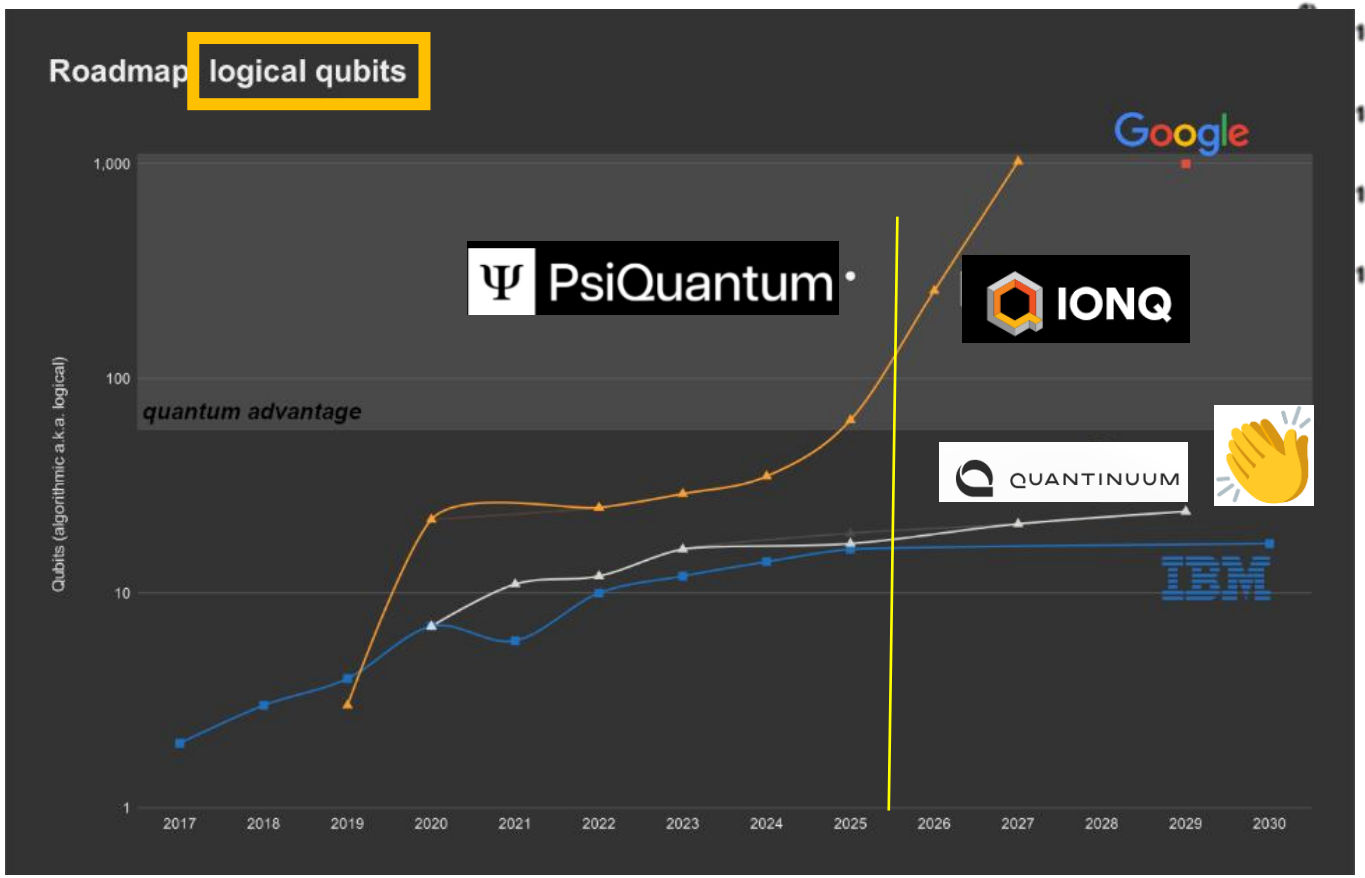
powerful tools for exploring

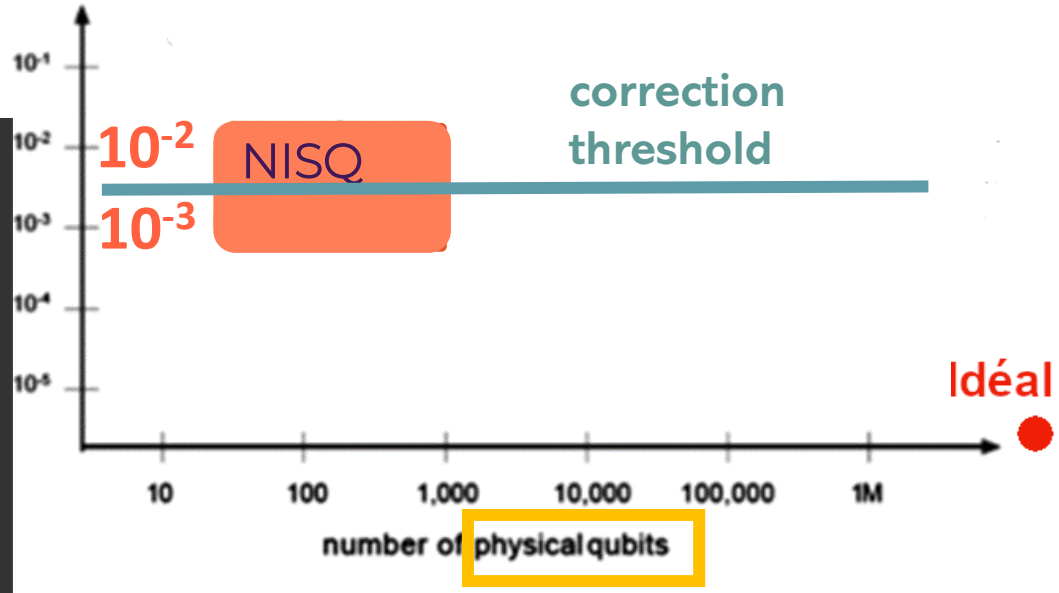
- Number of qubits



Error correction (cf classic)
Increase quality (correct errors by making the same errors…)
Threshold crossed

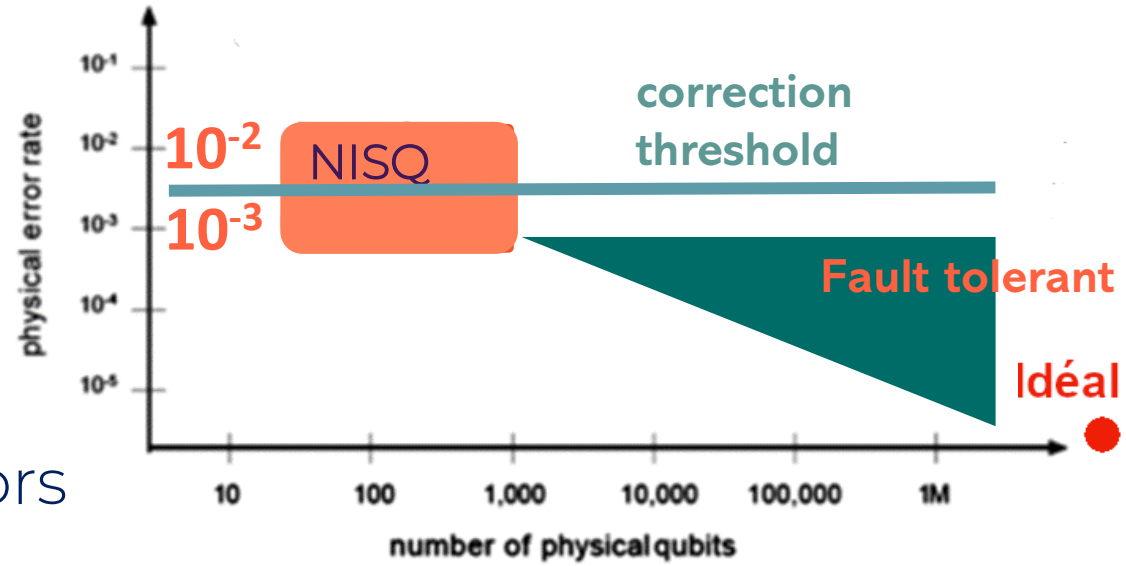
- Number of qubits





Psiquantum photons
IOnq trapped ions
Quantinuum trapped ions
IBM Supraconductor

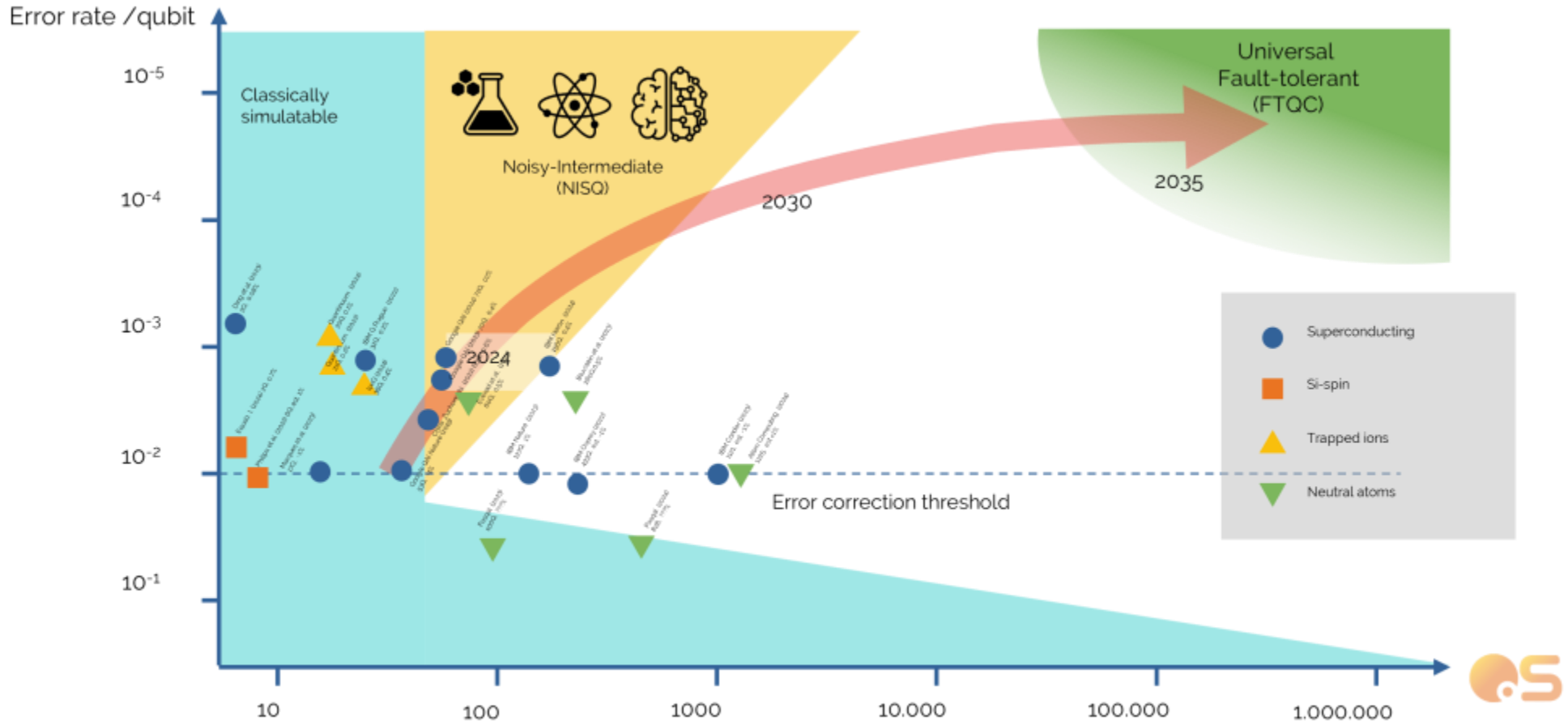
- Number of qubits



Error correction
focuses on detecting and correcting errors

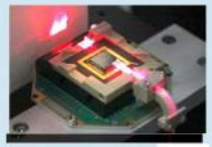
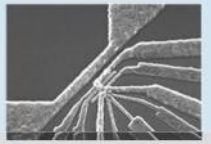
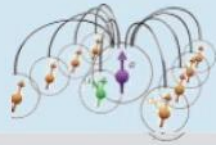
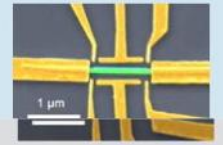
FTQC Fault Tolerant Quantum Computer
ensures that these corrections can be applied reliably without introducing new errors.

Allows quantum computations to run for extended periods



https://orangeqs.com/news/quantum_industry_roadmaps_2024/

Useful industrial applications need : some **hundreds of logical** qubits

	Superconducting	Photonic	Cold Atoms	Trapped Ions	Si Quantum Dots	NV-Center	Topological
							
Companies	Google IBM amazon	XANADU PsiQuantum	PASQAL atom computing QuEra Computing Inc.	QUANTINUUM IONQ oxford ionics	intel QuTech	QUANTUM BRILLIANCE	Microsoft
Number of Physical Qubits	>1000	>100	>1000	>50	>10	>10	n/a
Coherence Time	Transmon: ~500 us Fluxonium: ~1 ms	Time of photon flight	ms – s regime	> 10 mins	ms regime	s (cryo) ms (RT)	n/a
Gate Fidelities	Single: 99.99% Two: 99.9%	One: 99.9% Two: 99.5%	Single: 99.9% Two: 99.5%	Single: > 99.99% Two: 99.97%	Single: 99.9 % Two: 99 %	Single: 99.999% Two: 99.93%	
Gate Time	10 – 100 ns	1 – 100 ns	10 ns – 1us	1 – 100 us	0.1 – 10 us	~ 100 ns	n/a
Scalability	3D packaging Uniformity Optical link to >1E4	Large scale PIC with low loss	PIC for scalable control Optical link to >1E5	PIC & IC for scalable control Optical link to >1E3	3D packaging & cryoCMOS Yield & uniformity	C13 isotope growing random Optical link to > 1E2	n/a
Pros	Highly researched Engineerable qubits Market mature	Leverage telecom industry Network scaling	Large modules No mK cryogenics High connectivity	High gate fidelity No mK cryogenics High connectivity	Using CMOS module & expertise	Long coherence Quantum networking	Inherent fault tolerant properties
Cons	mK cryogenics Wiring bottleneck Network scaling	Large overhead Photon loss errors	New technology Long preparation times	Slow gate time Small modules	Strict material requirements Wiring bottle neck	Depending on random effects Small modules	Need proof-of-concept demonstration

Quantum Computing -- An Update



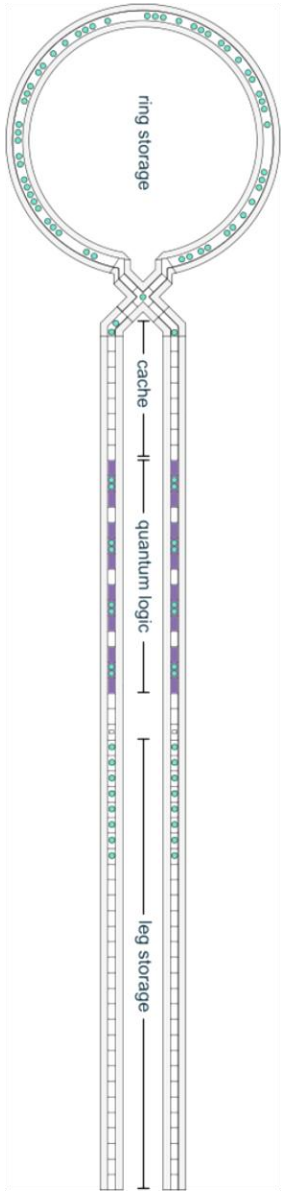
STEVE BLANK
OCT 22, 2024

Source: Applied Materials, industry data

Development roadmap

QUANTINUUM

Trapped ions



	2020	2023	2025	2027	2029
SYSTEMS:	H1	H2	QUANTINUUM HELIOS	QUANTINUUM SOL	QUANTINUUM APOLLO
PHYSICAL QUBITS:	20	56	96	192	1000's
PHYSICAL 2-QUBIT GATE ERROR:	1×10^{-3}	1×10^{-3}	$< 5 \times 10^{-4}$	$< 2 \times 10^{-4}$	1×10^{-4}
LOGICAL QUBITS:		10+	~ 50	~ 100	100's
LOGICAL ERROR RATES:		1×10^{-3}	$< 10^{-4}$	$\sim 10^{-5}$	1×10^{-5} to $1 \times 10^{-10*}$

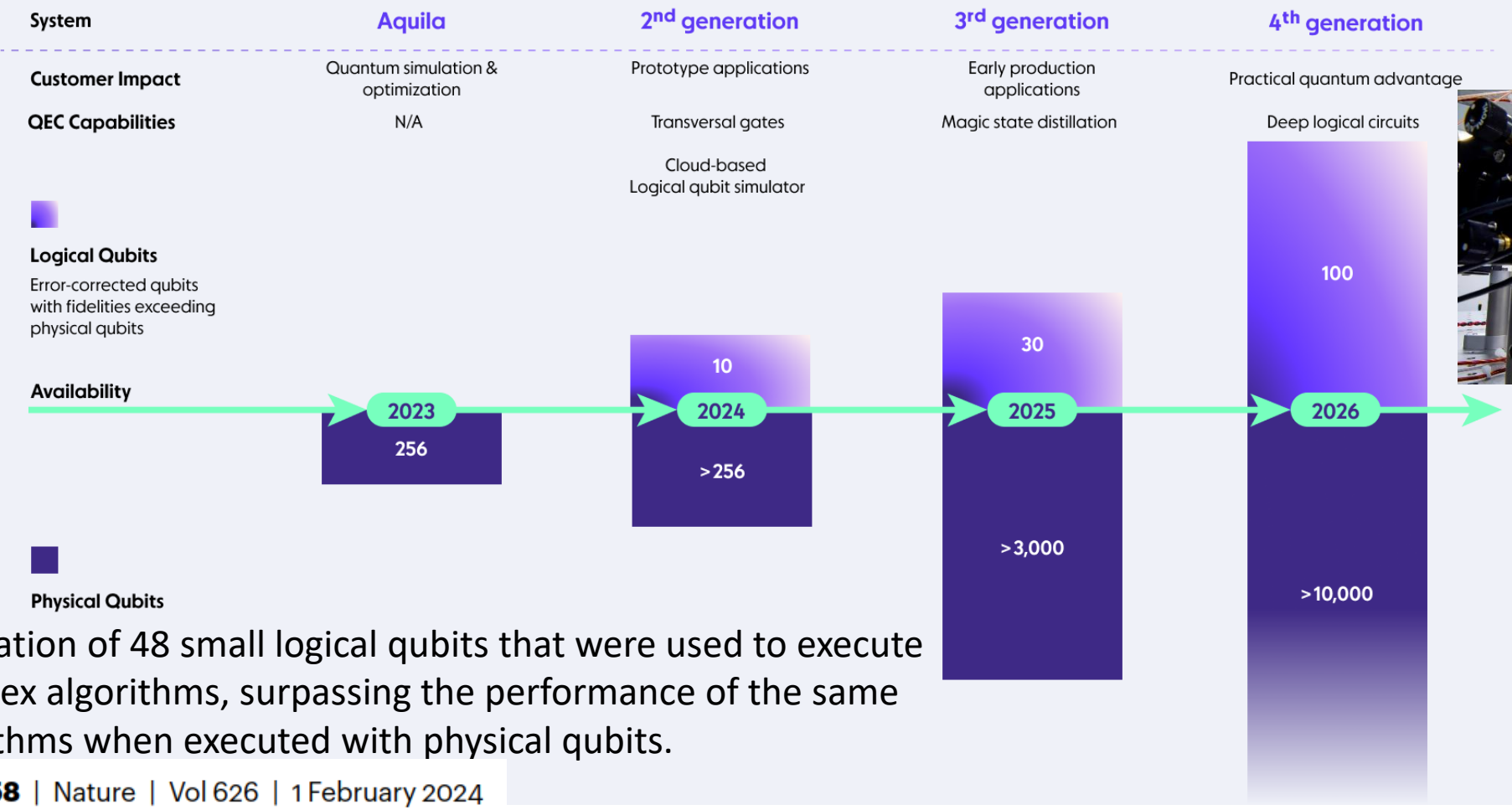
Q2 : 23 en 2025

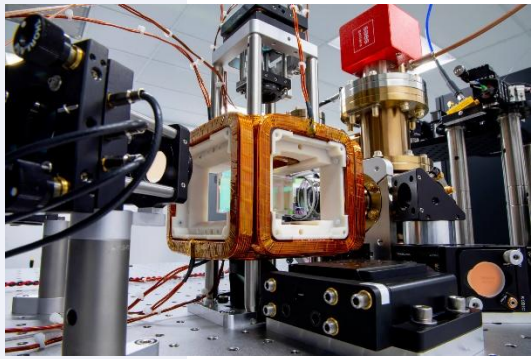
Q4 98 50 (November 2025)

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*alysis based on recent literature in new, novel error correcting codes predict that error could be as low as 1E-10 in Apollo (ref: arXiv:2403.16054, arXiv:2308.07915)

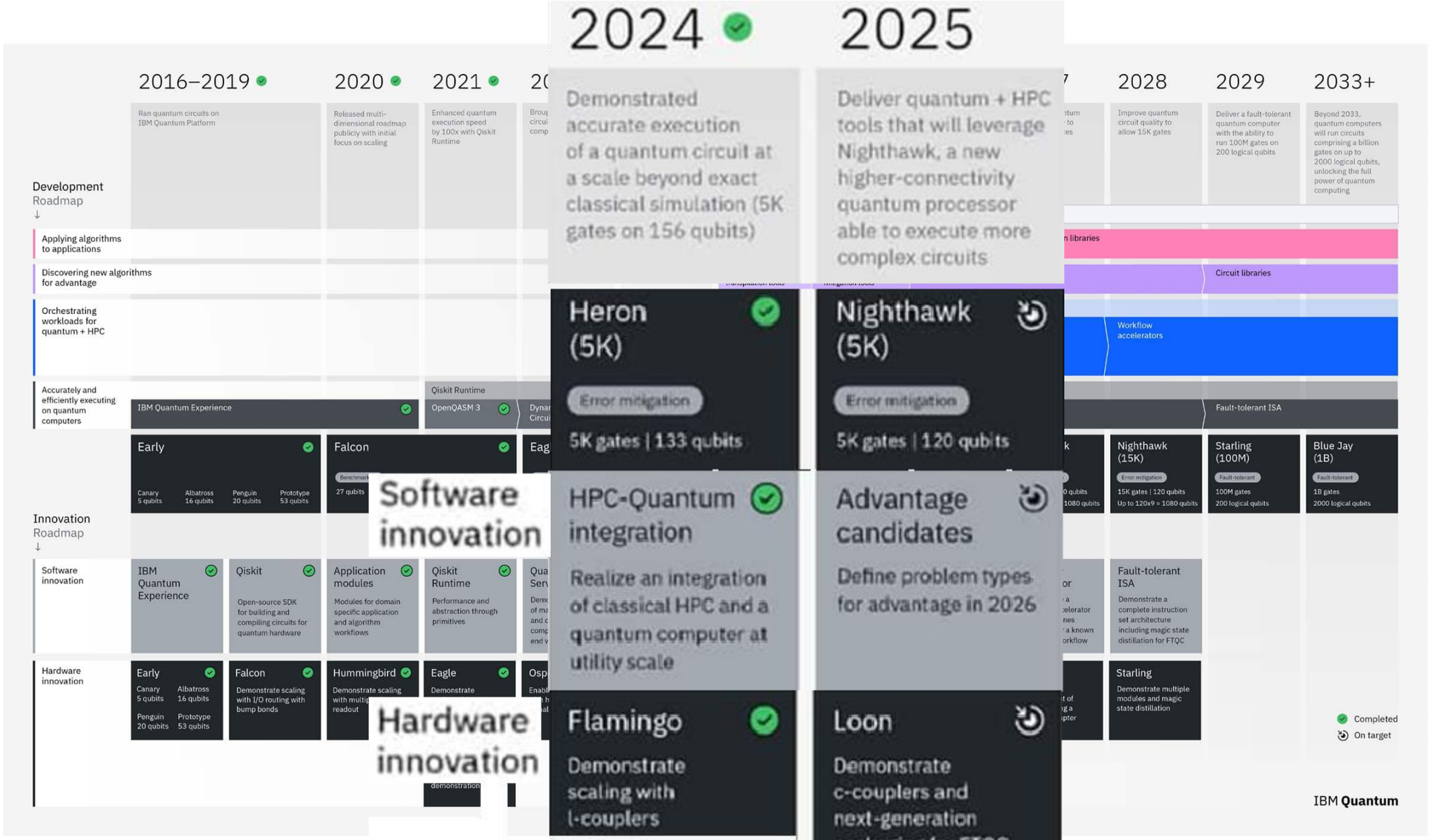
Error-Corrected Quantum Computing Roadmap



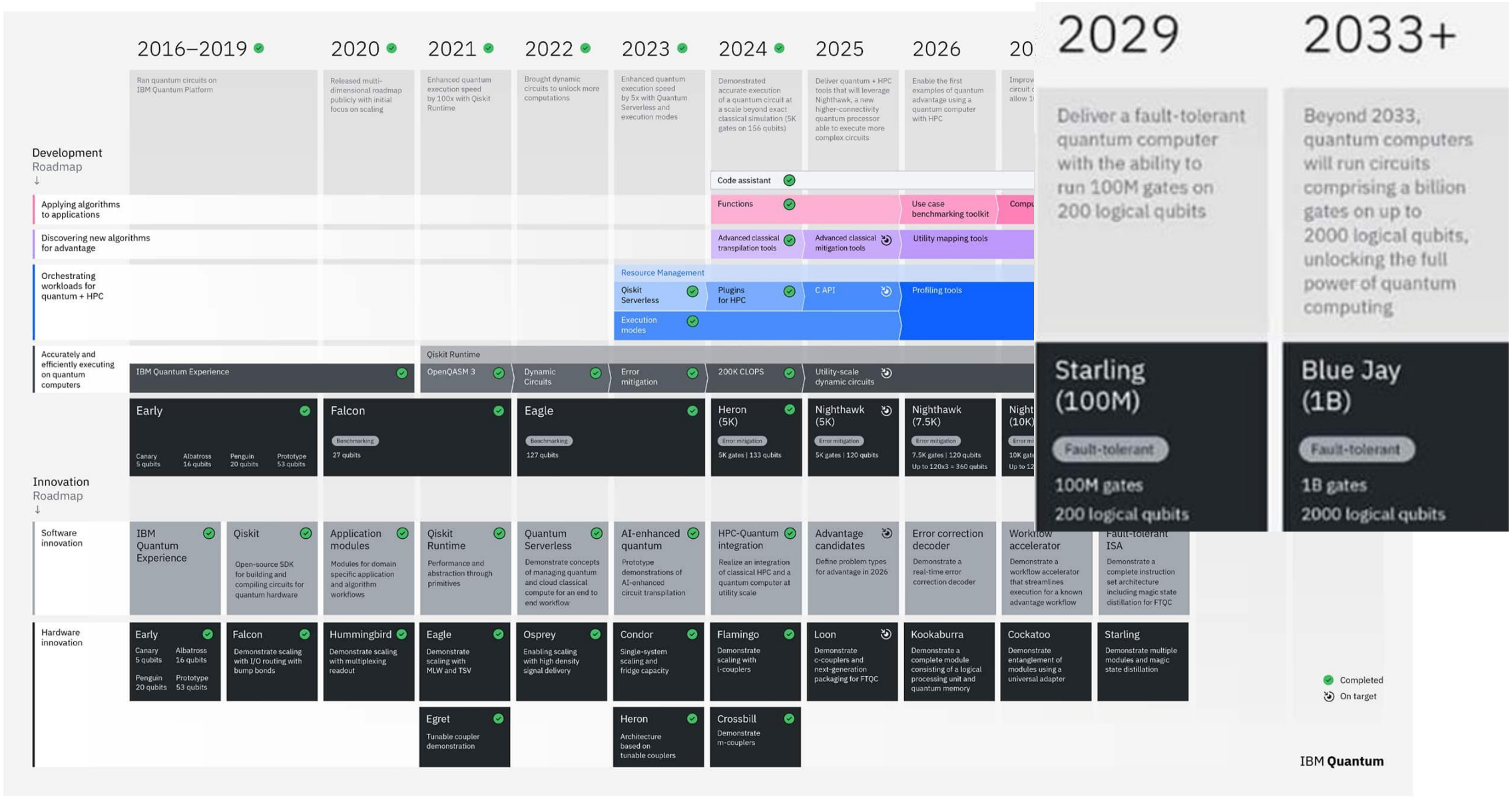


Realization of 48 small logical qubits that were used to execute complex algorithms, surpassing the performance of the same algorithms when executed with physical qubits.

58 | Nature | Vol 626 | 1 February 2024



2025 IBM Quantum Roadmap

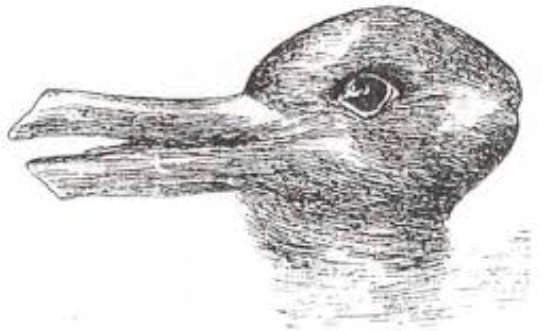


2025 IBM Quantum Roadmap.



Computer programs

(New paradigm)



Cf Thomas KHUN

« Gates» digital

Change qubits state

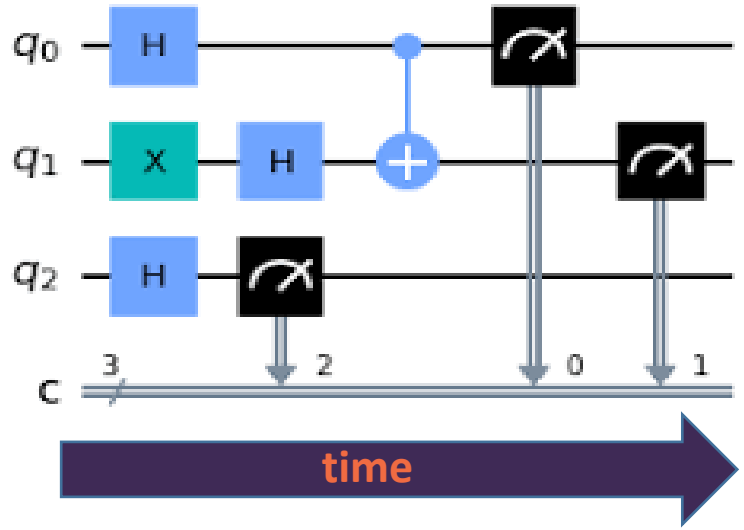


CIRCUIT

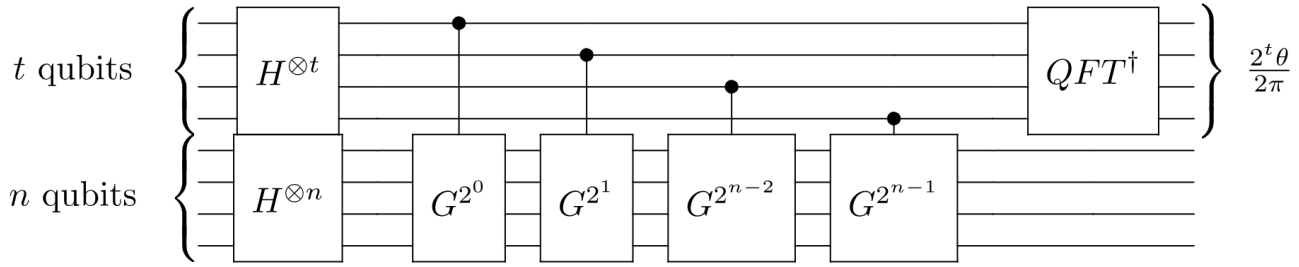
Superposition

Entanglement

Measure



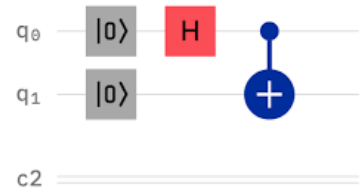
Decoherence



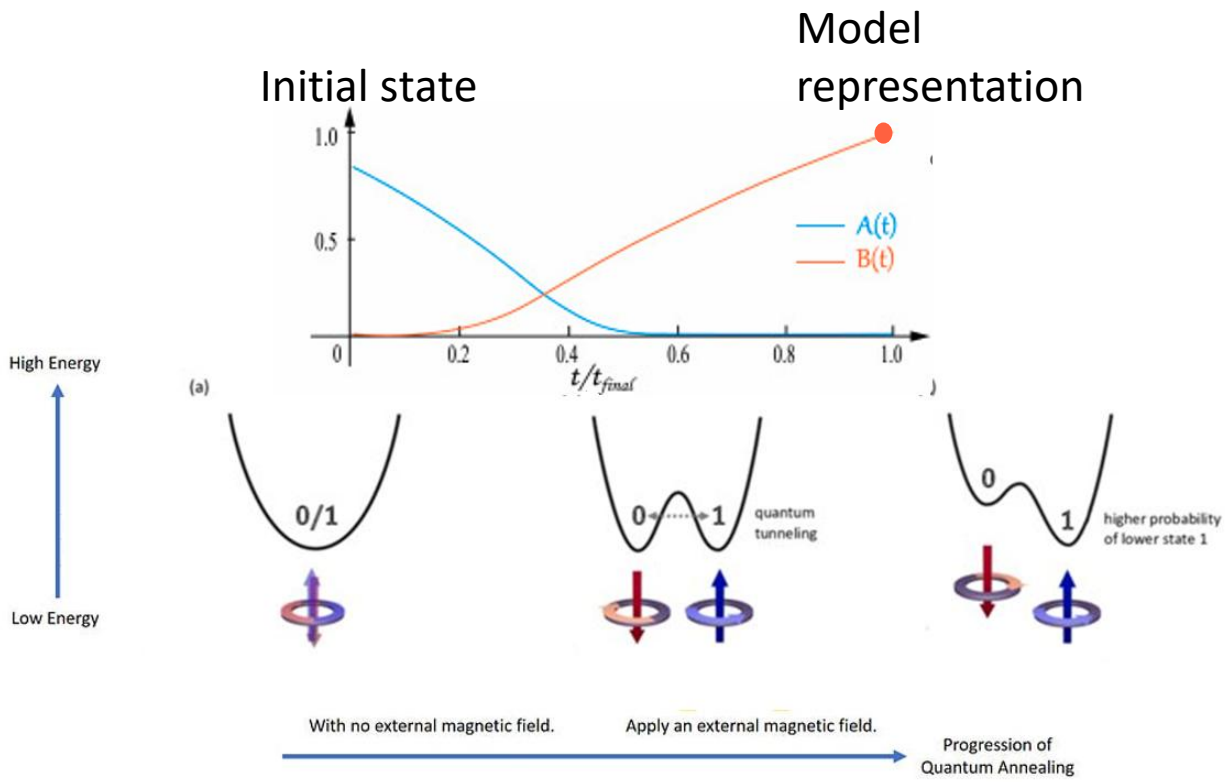
« Clasical » program Python, and so on…

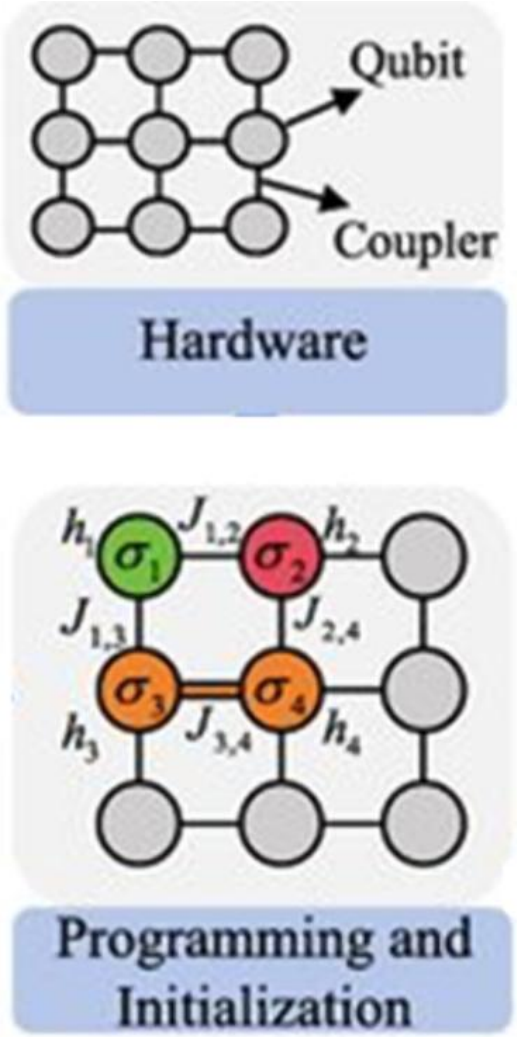
```
In [9]: 1 bell_circuit.code()
executed in 5ms, finished 12:37:57 2019-07-17

qr0 = QuantumRegister(2, 'q')
cr0 = ClassicalRegister(2, 'c')
qc = QuantumCircuit(qr0, cr0)
qc.h(qr0[0])
qc.cx(qr0[0], qr0[1])
qc.measure(qr0, cr0)
```



« Annealing » analogic





Everything happens before the calculation is made, **upstream** in the creation of the model.

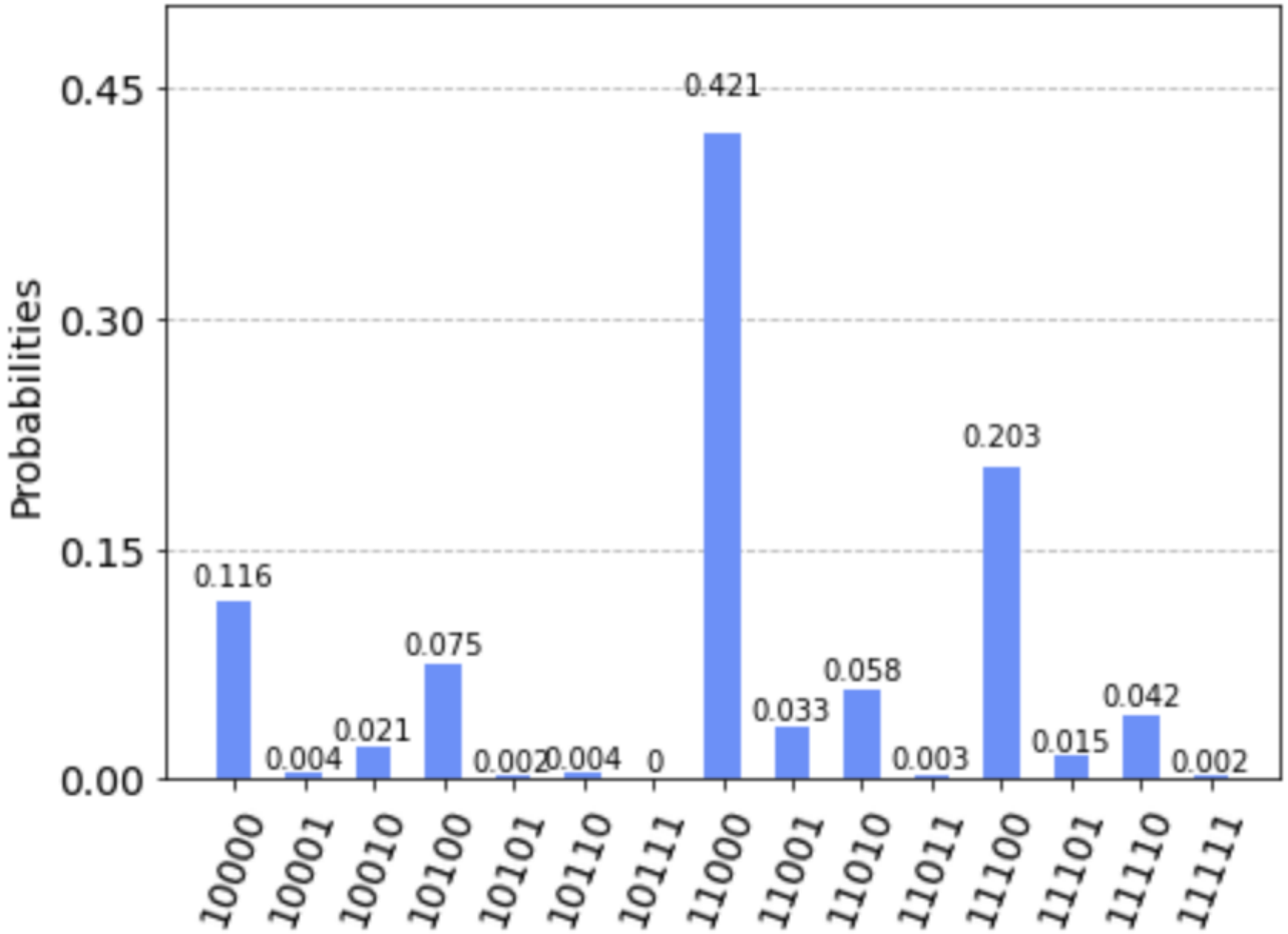
Résults

« probabilistic » mechanics

Mesure	Résultat	Probabiliste
	0	Avec 30% de chance
	1	Avec 70% de chance

3+2=5 à 95% =4 à 3% =6 à 2%

Perform the calculation a statistically significant number of times.

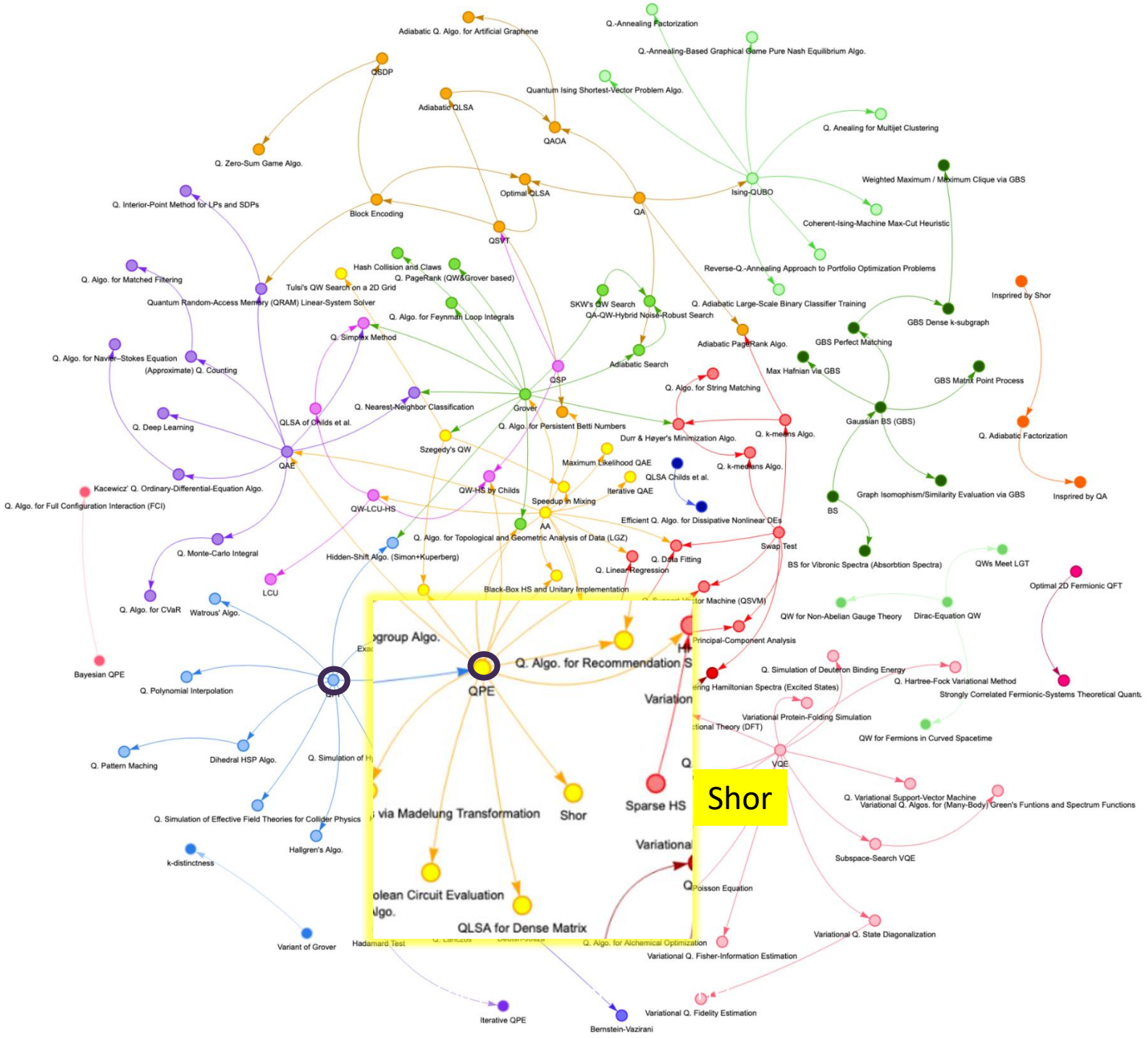


Quantum zoo



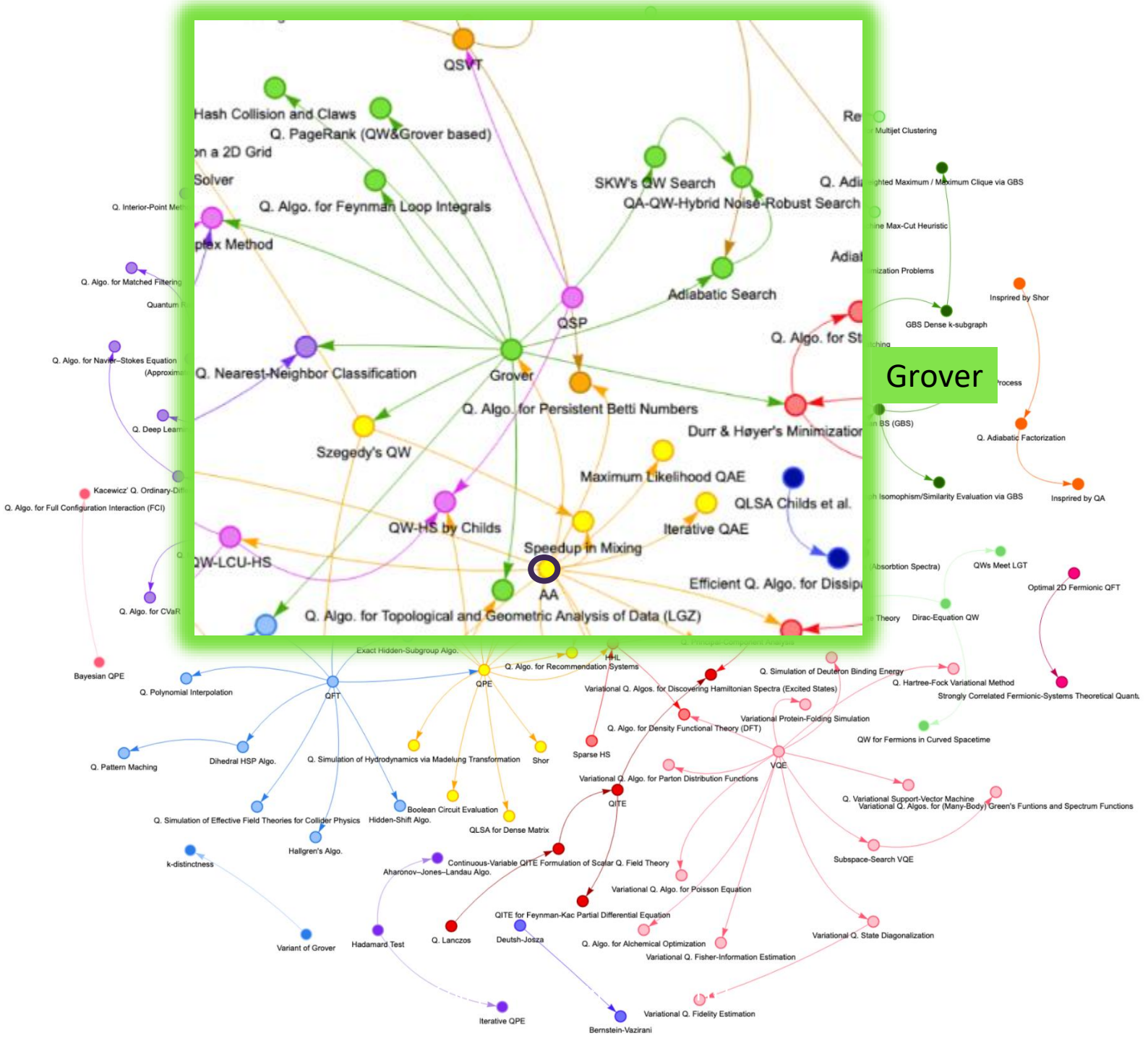
arXiv:2407.05178v1
[quant-ph] 6 Jul 2024

Quantum zoo



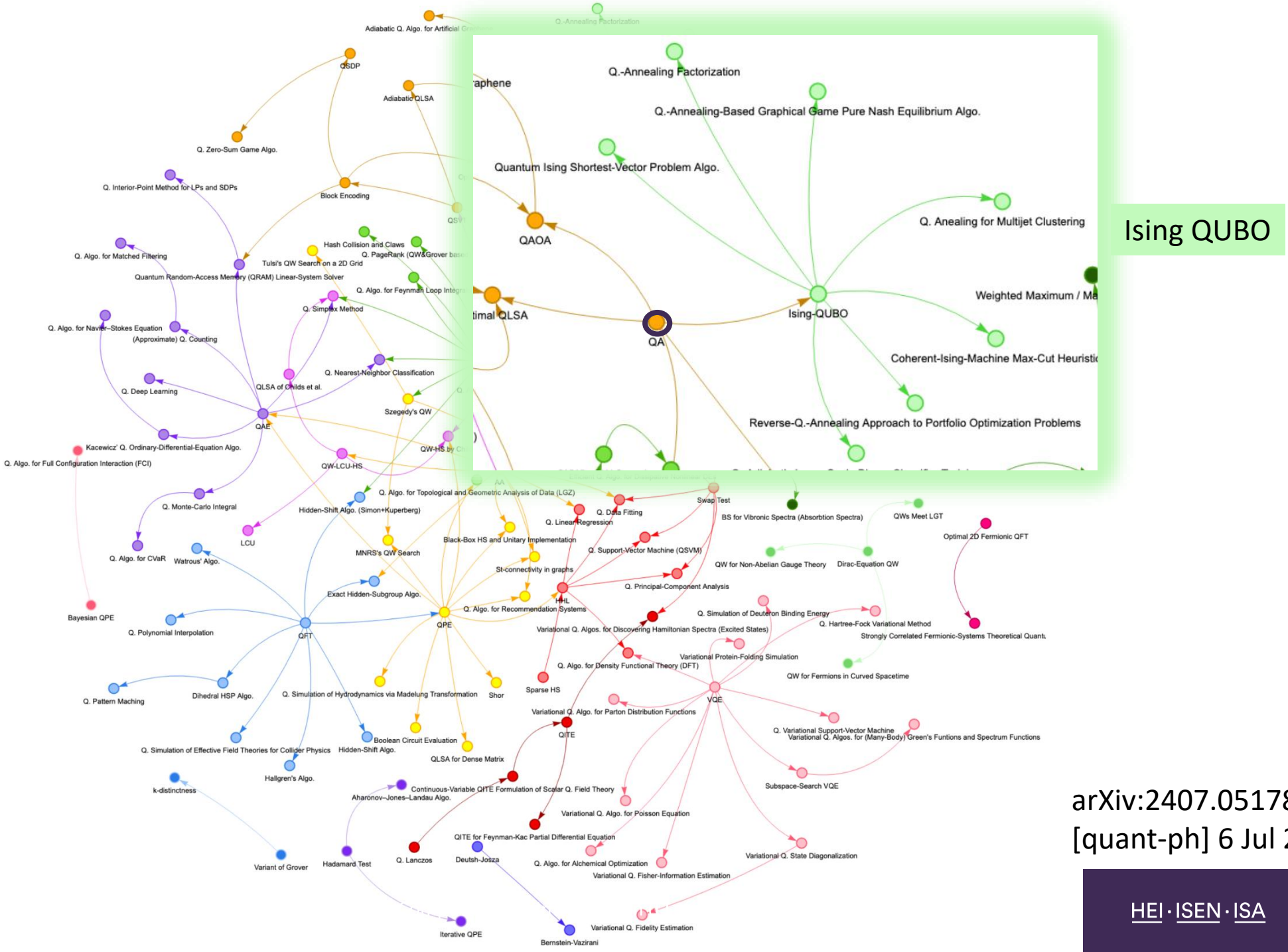
arXiv:2407.05178v1
[quant-ph] 6 Jul 2024

Quantum zoo



arXiv:2407.05178v1
[quant-ph] 6 Jul 2024

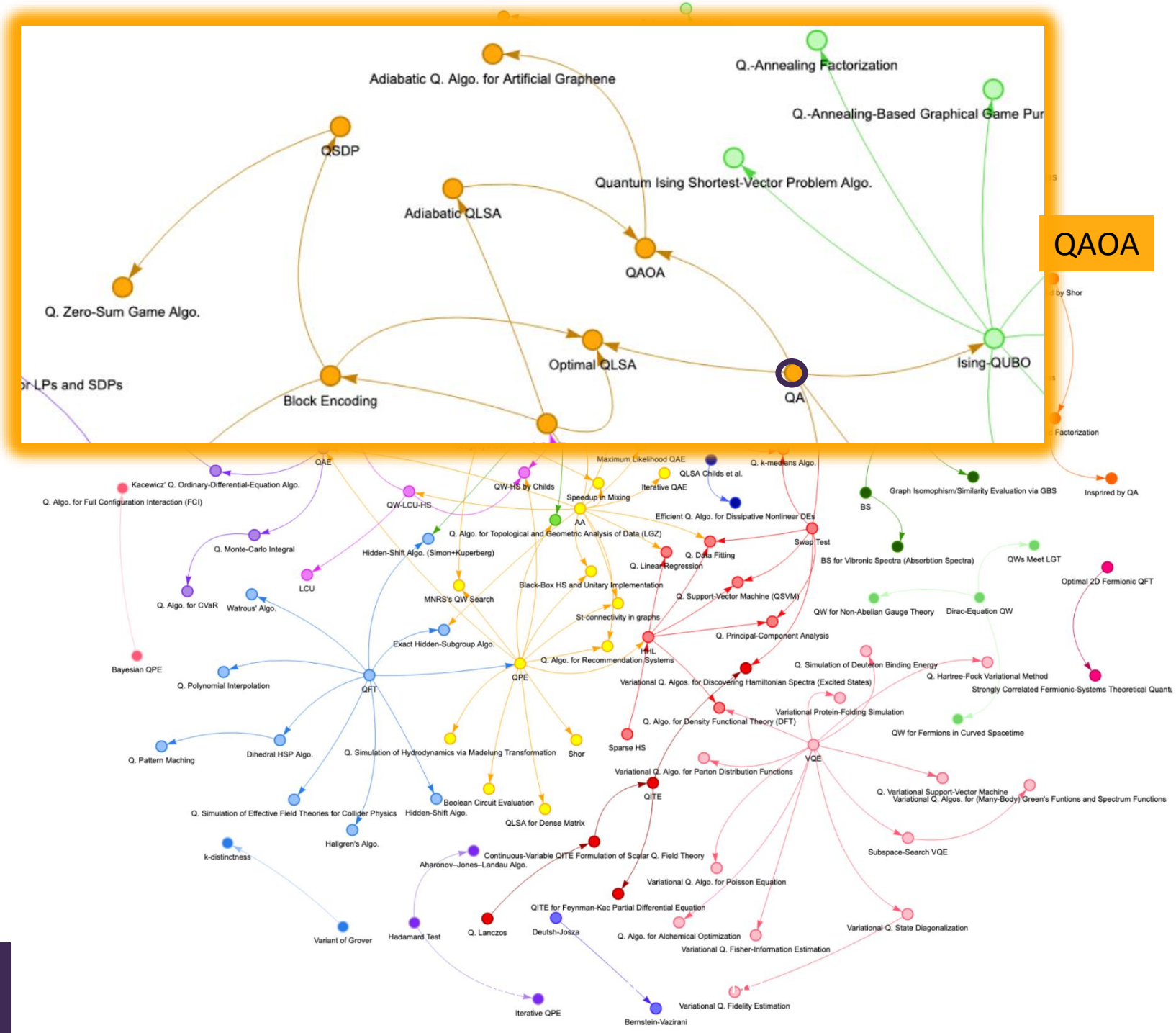
Quantum zoo



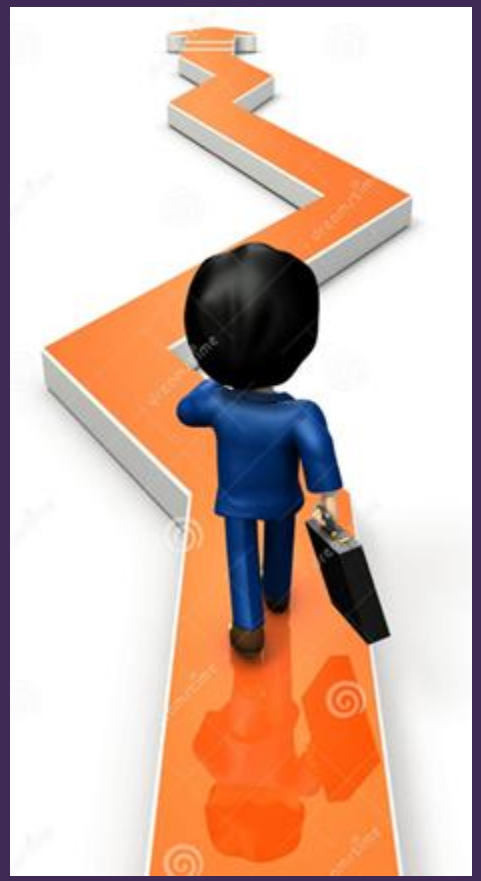
Ising QUBO

arXiv:2407.05178v1
[quant-ph] 6 Jul 2024

Quantum zoo



arXiv:2407.05178v1
[quant-ph] 6 Jul 2024



Perspectives

Enabling technologies

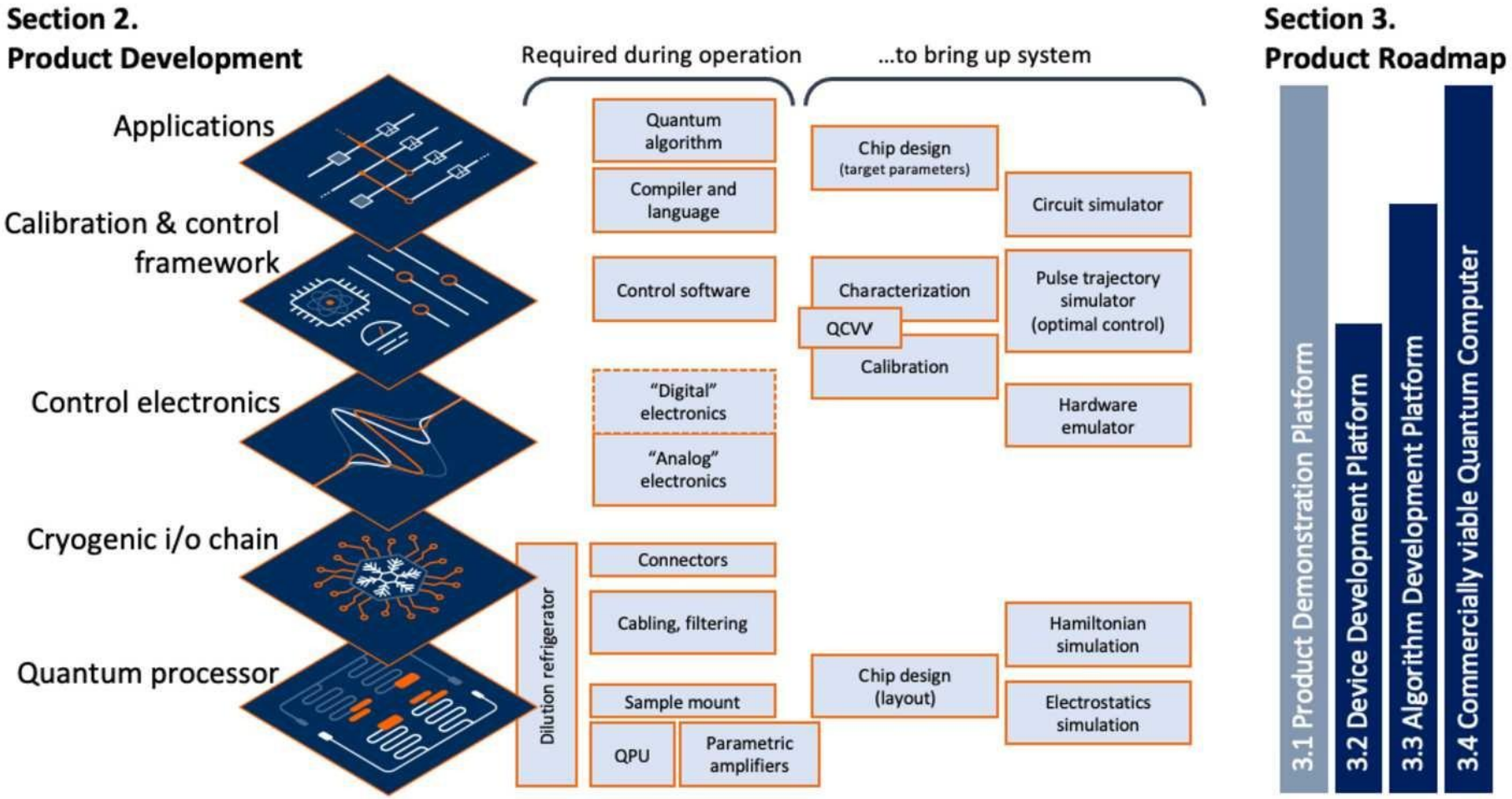




Figure 516: a market map of key enabling technology vendors. (cc) Olivier Ezratty, 2020-2024.

Hybrid computation

QPU - quantum card
in the same way as a graphics card

JÜLICH Forschungszentrum | JUNIQ QUANTUM USER FACILITY



Hybrid computation

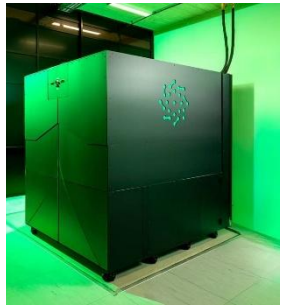


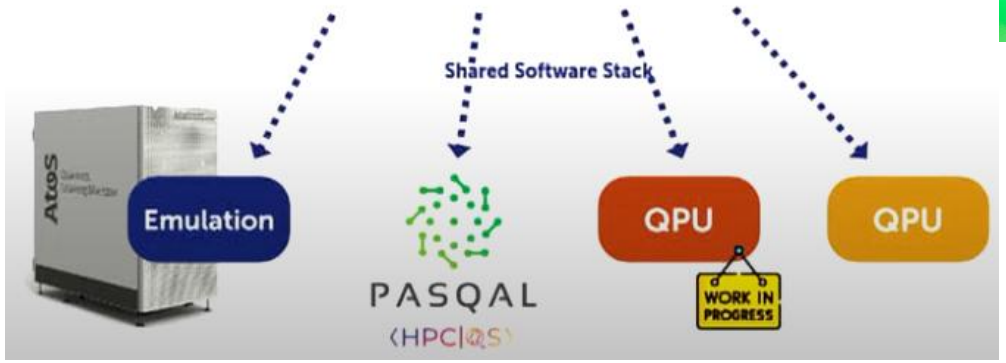
04/01/22

FRANCE HYBRID HPCQUANTUM INITIATIVE

Joliot-Curie

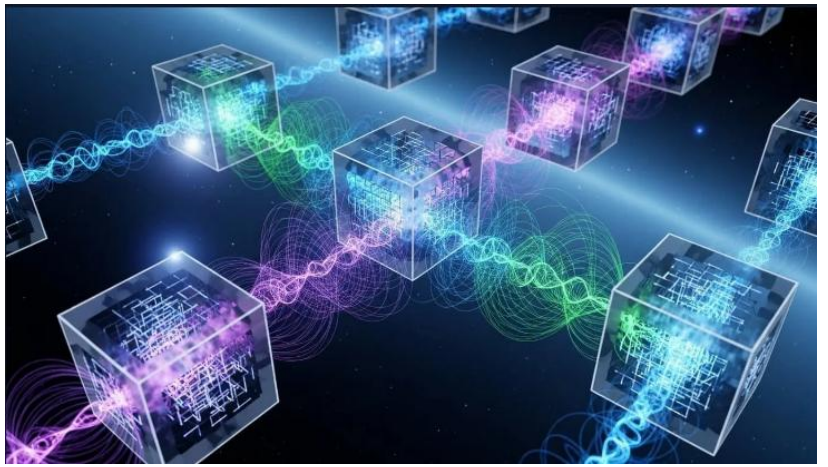




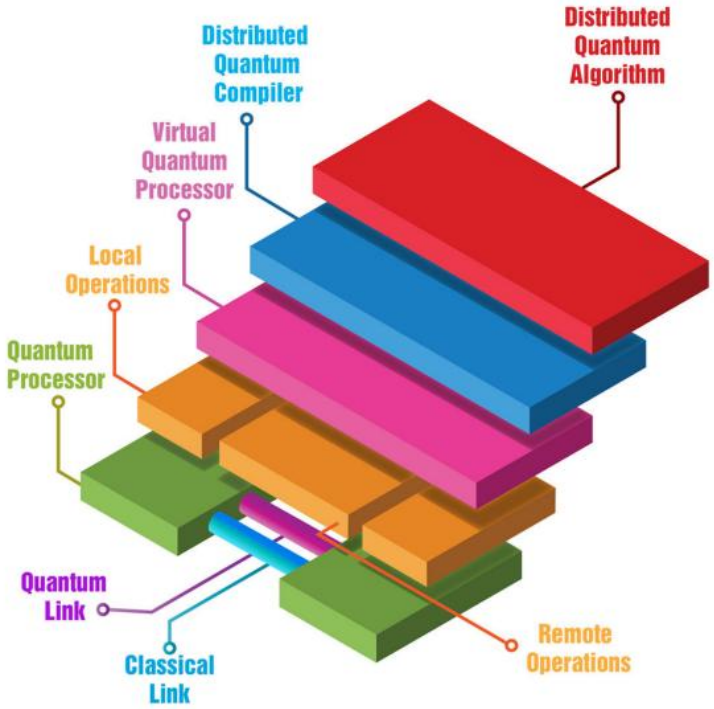


13/11/25 Ruby 100 qubits

Distributed Quantum Computing



- Execution of quantum computation across multiple QPU
- Sort of networking
- Path to large scale



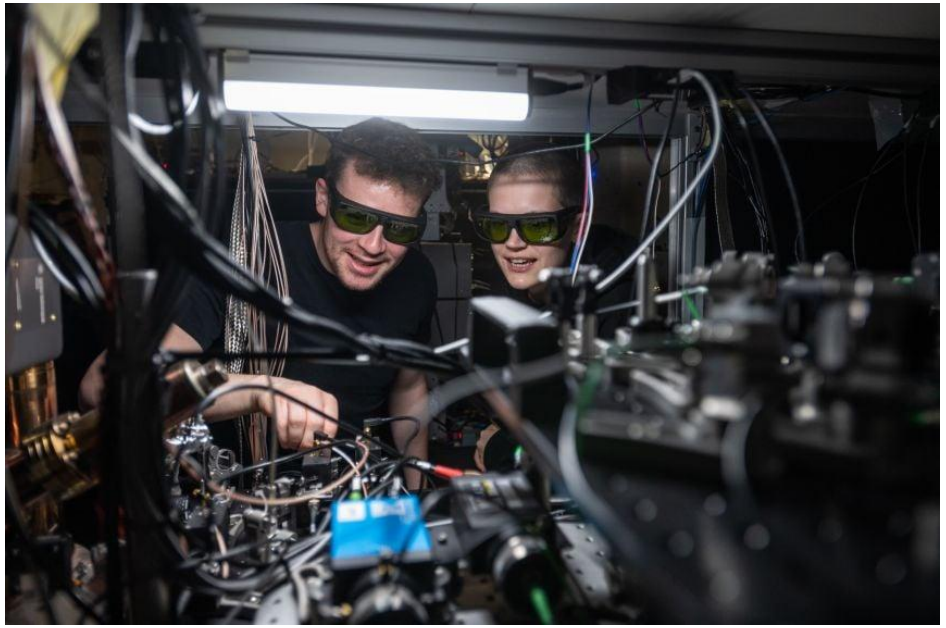
Distributed Quantum Computing

Article Nature | Vol 638 | 13 February 2025 | 383

Distributed quantum computing across an optical network link

D. Main[1]✉, P. Drmota[1], D. P. Nadlinger[1], E. M. Ainley[1], A. Agrawal[1], B. C. Nichol[1], R. Srinivas[1], G. Araneda[1] & D. M. Lucas[1]

[1]Clarendon Laboratory, Department of Physics, University of Oxford, Oxford, UK.



Dougal Main and Beth Nichol working on the distributed quantum computer. Image credit: John Cairns.

Between two photonically interconnected trapped-ion modules.
The modules, separated by about two metres
Heralded remote entanglement between the network qubits,
Teleport a controlled-Z (CZ) gate between two circuit qubits in separate modules
Distributed iSWAP and SWAP circuits (two-qubit operations).



What's the point of all this?

Use cases

"Quantum Computing: Business Use Case Maturity Horizons"

	HORIZON 1	HORIZON 2	HORIZON 3
INDUSTRIES	● Surfactants, Catalysts	▲ Oil shipping / trucking	◆ Seismic imaging
Chemical & Petroleum	● Chemical product design	▲ Refining processes	
Distribution & logistics		▲ Feedstock to product	
Financial services		◆ Drilling locations	
Health care & Life science	▲ Vehicule routing	◆ Freight forecasting	◆ Consumer Offer recommender
Manufacturing	▲ Network optimization	◆ Irregular behaviors (Ops)	
		■ Disruption management	
		▲ Distribution supply chain	
	◆ Credit / Asset scoring	■ Derivatives pricing	◆ Finance offer recommender
ALGORITHMES FAMILY	▲ Transaction settlement	◆ Irregular behaviores (Fraud)	
● Chemical simulation	▲ Portfolio management	■ Investment risk analysis	
■ Scenario simulation	● Drug discovery	◆ Accelerated Diagnosis	■ Disease risk prediction
▲ Optimization	● Protein structure prediction	◆ Clinical trial enhancement	
◆ AI / ML		◆ Genomic analysis	
		▲ Medical / drug supply chain	
Linear algebra		◆ Quality Control	◆ Structural design & fluid dynamics
Harrow–Hassidim–Lloyd (HHL) algorithm	● Quantum Chemistry	▲ Process planning	
	● Material discovery	▲ Manufacturing supply	
		▲ Fabrication optimization	

3 TIMES

Inspiré d'IBM 2019

Roadmap



Proof of concept:

power plant maintenance scheduling, product portfolio optimization, train scheduling

Medium size problem :

portfolio optimiszation, traveling salesperson and graph coloring problems

Advances logistics and routing, energy grid optimization, risk management, infrastructure planning, telecom and data center management

https://meetiqm.com/technology/applications-algorithms/

Industry and quantum computing

industry is engaging quantum technologies for
real-world research and potential applications

Some examples

IBM Q

- **JPMorgan Chase** —*financial problems*, including portfolio optimization, risk analysis, and pricing.
- **Daimler AG** — *automotive manufacturing, materials science, and optimization problems*

Quantinuum

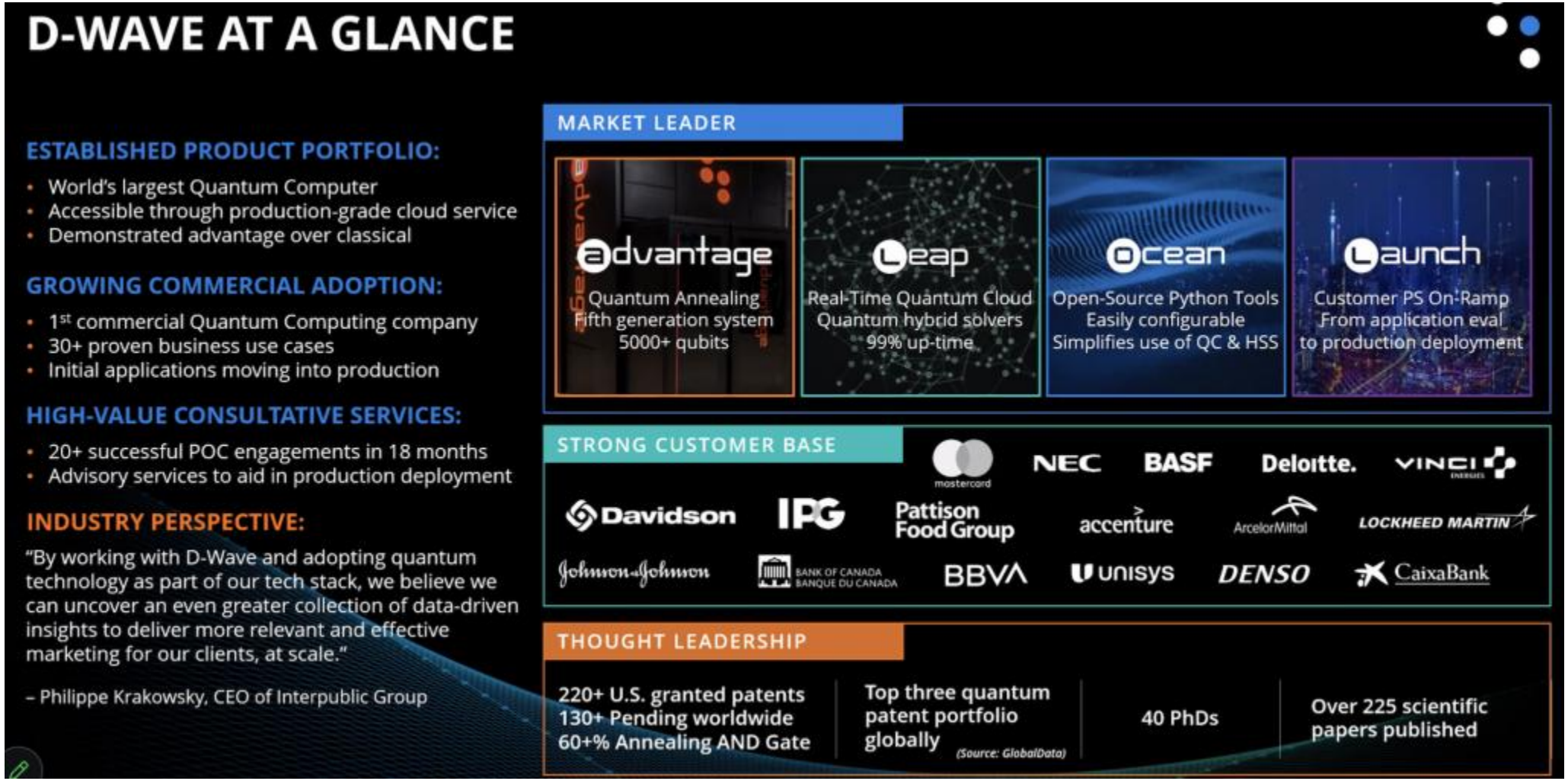
- **BMW Group** — *quantum material research* for sustainable mobility, e.g., fuel cell catalyst design.
- **Amgen** — *life sciences* with hybrid quantum–machine learning approaches.

PASQAL

- **LG Electronics** —quantum algorithms for industrial challenges (e.g., multiphysics simulation, optimization, material discovery) / core technologies for neutral-atom quantum computing infrastructure.
- **Eni (Italian energy company)** — develop quantum computing solutions for the energy sector

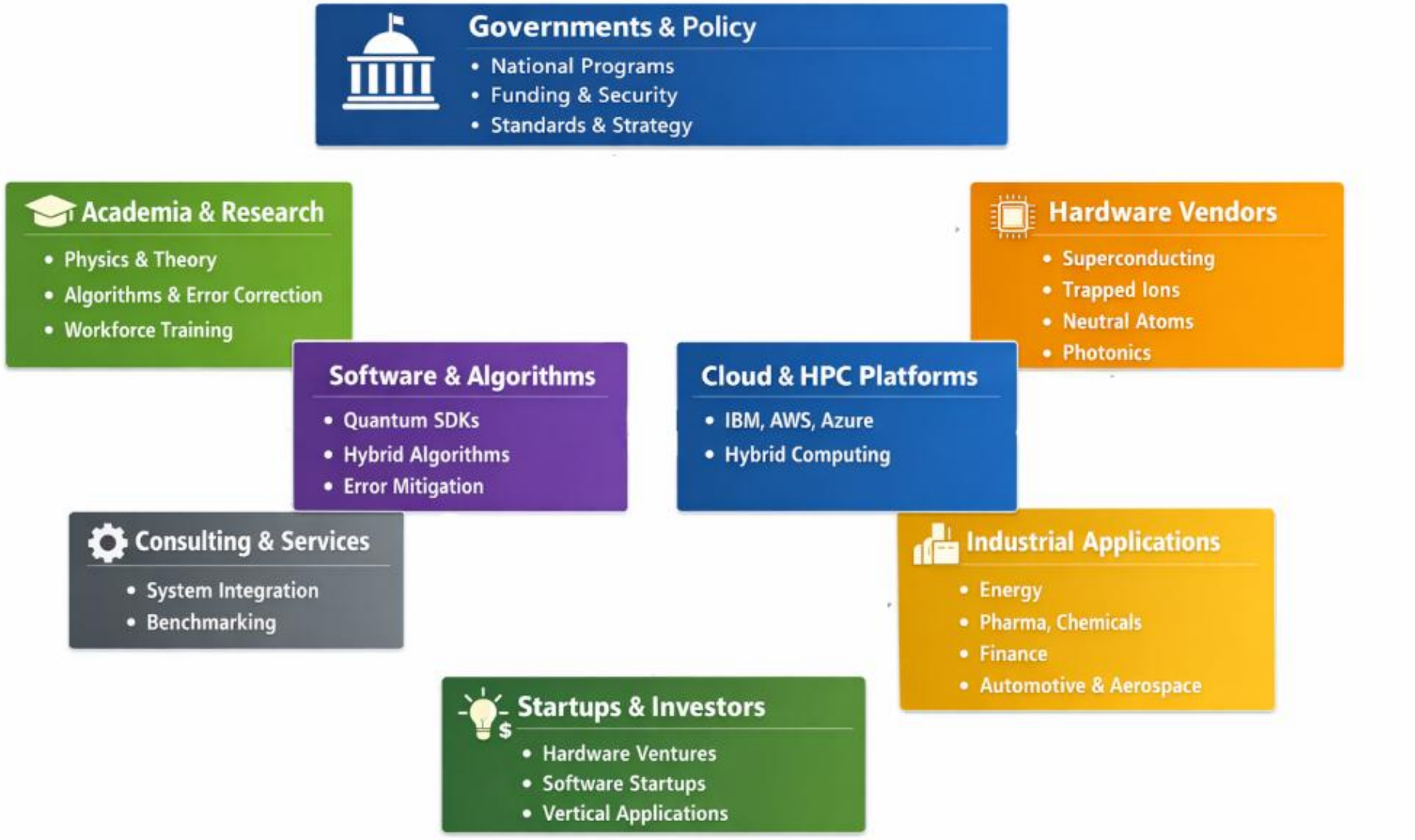
https://www.pasqal.com/blog/content/use-cases/

https://www.dwavequantum.com/learn/featured-applications/?page=7



TO CONCLUDE





The Quantum Computing Ecosystem

« Quantum computer »

Keep in mind : Only adapted to particular problems

Be specific :

* Analogic / NSQ / FTQC
* Qubit technology
* physical / logical
* Error rate / correction
* Decoherence time
* Which algorithm
* Alone? Hybrid? Distributed?



THANK YOU

for your attention!

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